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# Extrapolative households and strategic firms: Evidence from China's land and housing market☆

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ABSTRACT

This paper examines households' extrapolative behaviors and developers' strategic responses in China's land and housing markets. Our stylized model predicts that when households extrapolate in the housing market, developers with nearby land holdings strategically bid higher, anticipating rising housing prices to boost the value of their existing stock. Using the exogenous timing of land auctions, we show that higher land premiums increase transaction prices and volumes of nearby houses. Furthermore, developers with nearby land stocks consistently pay higher premiums in the land auctions. We exploit a policy reform and implement various tests to support the hypothesis that developers strategically influence homebuyers' expectations.

## 1. Introduction

In markets characterized by information asymmetry, participants actively seek and interpret signals to guide their decisions. This dynamic is particularly pronounced in the housing market, where households frequently rely on historical housing price trends to predict future values, a behavioral pattern commonly referred to as extrapolation (Glaeser and Nathanson, 2017; Barberis et al., 2018). The challenge with extrapolating from past price growth is that homebuyers may not be able to tell whether recent appreciation reflects stronger fundamentals or a temporary change in beliefs. This raises an important question: If homebuyers extrapolate, do more sophisticated market participants, such as real estate developers, strategically exploit this behavior to their advantage?

This paper investigates whether real estate developers in China use land auctions to shape housing prices. By influencing market signals, developers may affect homebuyers' expectations, thereby altering housing demand and pricing dynamics. Understanding developers' incentives to influence land auctions is crucial, as such behavior significantly impacts land and housing markets, with broad policy implications for urban development in other developing countries.

China provides a unique setting to examine these issues. First, all urban land in China is state-owned, with local governments leasing parcels to real estate developers through auctions. Prices of auctioned land, particularly record-breaking transactions, are widely publicized. The land price shocks help us empirically identify extrapolations in the housing market. Second, local governments face dual objectives: generating revenue from land sales and stabilizing housing prices in accordance with the central government's principle that *"houses are for living in, not for speculation"*. These conflicting goals have prompted policies aimed at anchoring homebuyers' price expectations. By analyzing whether developers' strategies in land auctions shift under conditions where they can no longer influence homebuyers' expectations, we can identify whether developers actively steer homebuyers' behaviors.

To guide our empirical analysis, we construct a stylized model to examine households' and developers' strategies in the housing and land markets. Central to our model is the assumption that when a land parcel is sold at a price higher than the locally predicted price, it triggers an increase in the demand for nearby houses. While our model does not directly specify the underlying mechanism of this assumption,

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Appendix A demonstrates that when nearby land prices positively influence households' expectations of future housing prices, the demand for local housing increases when the nearby land is auctioned at a high premium. Based on this assumption, we show that when developers possess additional housing stocks near the auctioned land parcel, they are willing to pay a higher price for the land compared to developers without nearby land assets. This group of developers benefits from: a cross-subsidy effect, where higher land premiums increase demand for the developer's existing housing units, and the exercise of enhanced local market power, which boosts the developer's profitability. Our concept of auction-based strategic signaling aligns with the theoretical model discussed in Benabou and Laroque (1992), describing how insiders with access to imperfect private information may use skewed public statements to influence stock prices.

We begin our empirical analysis by examining whether the assumption that households extrapolate from high land prices holds. To capture the information content of auction outcomes, we construct a land auction premium that measures how much the winning bid for a parcel exceeds the local housing price growth rate. For each auction, we link its land-price premium to housing transactions that occur within a 5 km radius. Homes located within 3 km of an auction form the treatment group, while those 3–5 km away serve as controls. Comparing the dynamics of transaction prices for homes inside and outside the 3 km radius around the auction date yields an event-study style Difference-in-Differences design. Our identification relies on the parallel-trend assumption: absent land auctions, housing prices for units near and far from auctioned parcels would have evolved similarly over event time, conditional on a rich set of fixed effects and controls. This assumption is plausible in our setting because the exact timing of land auctions is driven primarily by administrative and procedural considerations rather than short-run neighborhood-specific shocks to housing demand.

Before the land auction, the housing prices near the auctioned land were similar to those further away, and the differences were small and insignificant. The insignificant pre-trend validates our identification strategy. In the month of the auction, a one standard deviation increase in the land premium leads to a 1.7 percent change in the transaction price of existing houses located within 3 km of the land parcel relative to those located between 3 and 5 km. This response exhibits strong momentum, continuing for three months after the auction, after which the estimated effect attenuates. For newly-built houses, we find similar patterns: housing prices rise significantly following high-premium auctions and continue to increase for more than six months post-auction. Theoretical frameworks focusing on extrapolative households also predict increased transaction volumes following higher returns (Barberis et al., 2018; Liao et al., 2022; DeFusco et al., 2022). We find a significant increase in the number of nearby homes purchased during the auction month. The effect exhibits strong momentum.

To rule out the influence of unobserved land characteristics and heterogeneous treatment effects, we replicate our analysis using a quality-adjusted land premium and the estimator proposed by De Chaisemartin and D'Haultfœuille (2024). Our findings remain robust to this alternative measure and estimator: the transitory price surge, the asymmetry across market cycles, and the volume response all persist. This confirms that the observed spillovers stem from the price signal itself.

We interpret the concurrent rise in prices and transaction volumes as evidence that households update their beliefs about local housing prices after observing high land premiums. We assess this mechanism via two tests focusing on attention and expectation updating. First, to test if auction outcomes are sufficiently salient for households to process in real time, we analyze the Baidu Index (the Chinese counterpart to Google Trends). We find that a one-unit increase in the land premium corresponds to a 45% spike in search activity, suggesting that competitive auctions generate a proportional informational response. Second, we investigate whether high-premium auctions cause households to revise their expectations upward. Leveraging a unique survey eliciting respondents' beliefs about future housing prices, we provide direct evidence that the increased demand is driven by households becoming more optimistic about future housing prices. We find that households living near land parcels auctioned at a higher premium are more optimistic about both short-term and long-term housing prices, after controlling a rich set of individual and time fixed effects.

This extrapolation from land premiums by households incentivizes developers with housing stocks near the auctioned land to increase the land price. Developers that already hold land within a three-kilometer radius of a parcel on the auction block bid systematically higher: in the preferred specification, controlling for city-year variation, land-use type, and auction format, their winning bids carry a 3.9 percentage-point larger premium than those of non-incumbents. Two interaction tests show why. When the nearby-land indicator is interacted with (i) the log floor-area of the developer's nearby unsold housing inventory and (ii) the change in its local market share that the new parcel confers, both terms are positive and significant. Jointly, they imply that at sample means the extra payment averages 5.72 percent, and more than half of that comes from unsold stock. This suggests that developers "overpay" to increase nearby sale prices and cross-subsidize the units they still hold. Moreover, this premium materializes only in boom periods and disappears in downturns, confirming that the strategy pays off only when exuberant buyers are primed to extrapolate auction signals.

Our robustness exercises rule out the three non-strategic explanations for why incumbents bid more aggressively when they already own nearby land. First, the premium survives after removing all listed firms and the top-100 developers, showing that it is not driven by an ability to squeeze out smaller rivals. Second, developers pay the extra premium whether or not a new school or metro line is scheduled to open within three kilometers of the site, so privileged knowledge of future infrastructure cannot be the driver. Third, the effect vanishes once a developer's local inventory has been fully sold: substituting *sold* for *unsold* stock in the interaction term yields an economically and statistically insignificant coefficient. A quasi-experimental policy test clinches the case. In cities that adopted "limit-house-price, compete-land-price" auctions, where the sale price of the completed homes is capped, the nearby-land premium collapses to zero, while it averages 6.6 percent in unconstrained cities. Event-study patterns indicate that this divergence emerges around policy implementation rather than gradually over time.

Combining the robustness checks yields a coherent pattern. The strategic premium appears only when it can pay off: it survives a battery of falsification checks, vanishes once the developer's local inventory is sold, and is present exclusively in boom phases of the housing cycle, when households' extrapolative demand is strongest. Together, these patterns point to a cross-subsidy channel: developers exploit exuberant markets by using land bids to lift the prices of their existing nearby holdings.

This paper makes two contributions. We are the first, to our knowledge, to empirically identify developers' strategic use of land auctions to influence housing demand. Our findings complement the theoretical literature on extrapolative households in housing markets. Glaeser and Nathanson (2017) introduced naive homebuyers, who neglect to account that previous buyers were learning from prices and instead take past prices as direct measures of demand. They find that when all agents follow their behavioral theory, their calibrated model closely matches the empirical value of one- and two-period auto-correlations in housing prices. Other works in this vein include Barberis et al. (2018), Burnside et al. (2016), DeFusco et al. (2022), Guren (2018), and Piazzesi and Schneider (2009). Despite the theoretical success, empirical work on extrapolative homebuyers is scarce. Armona et al. (2019) is a rare exception. They investigated how consumers' home price expectations respond to past home price growth and found that year-ahead home price expectations are revised in a way consistent with short-term momentum in home price growth. In contrast, we identify homebuyers' extrapolative housing demand without eliciting their subjective beliefs by examining how they respond to unexpected price movements from land auctions. Chan et al. (2016) compared homeowners' house valuations and market estimates and found that homeowners were reluctant to promptly adjust their valuations downwards in market downturns. Our findings suggest that during market downturns, homebuyers are less responsive to land auction premiums, thereby contributing to the existing literature by providing empirical evidence derived from quasi-experimental variations.

Our second contribution comes in expanding the focus of the literature studying China's land market. We do so by focusing on the households' extrapolation from land prices and developers' strategic responses. Existing studies have studied China's land market primarily from the political economy perspective. For example, there is ample evidence linking land revenue with corruption (Cai et al., 2013; Chen and Kung, 2016; Chen and Kung, 2019; Chen et al., 2023). Du and Peiser (2014) examined local governments' land hoarding behavior. Chang et al. (2025) is closely related, and they shared our intuition that land prices shape residents' and firms' expectations and confidence regarding the local economy. However, they focused on local governments' active management of land prices after the Covid-19 pandemic shock.

The rest of the paper is organized as follows. We describe the institutional background and data in Section 2. Section 3 describes the conceptual framework, which provides hypotheses for the empirical analysis in Sections 4 and 5. Section 6 concludes.

## 2. Background and data

### 2.1. Institutional background of China's land auctions

***Overview.*** Urban land auctions in Chinese cities follow a tightly structured, multi-stage process. At the beginning of each year, the municipal natural resources bureau leads the preparation of an annual land supply plan. In drawing up this plan, officials align the proposed land supply with the city's master plan, the broader economic development strategy, and the land-use quotas allocated by higher-level governments. The plan specifies the total quantity of land to be supplied over the year, its spatial distribution within the city, and its composition by use (residential, commercial, industrial, and so on).

Within this annual framework, bringing any specific parcel to market requires substantial preparatory work. The core tasks are handled by the local land reserve institution, which acquires land through expropriation or purchase and undertakes preliminary development. In practice, this involves providing basic infrastructure, such as water, electricity, roads, and other utilities, and leveling the site so that it is ready for immediate development once sold. In parallel, the natural resources bureau sets detailed planning conditions for the parcel, including land use, floor-area ratio, building density, and green-space ratio. Based on these conditions, the government commissions a professional appraisal of the land value and uses it to determine the reserve price.

Once these preparatory steps are complete, the parcel is formally introduced to the market. The natural resources bureau publishes an official transfer announcement, typically on the China Land Market website and the local public resources trading platform, specifying the parcel's location, area, planning requirements, starting price, required bidding deposit, application deadline, and the auction format. Only firms that pay the required bidding deposit within the prescribed period are qualified to participate. On the scheduled auction date, the auction is conducted according to the announced rules, and the parcel is transferred to the highest bidder subject to the reserve price and any binding policy constraints.

***Mapping to Empirical Design.*** Three features of this institutional environment are particularly important for our empirical analysis. First, land sales can be observed separately from housing transactions. Unlike

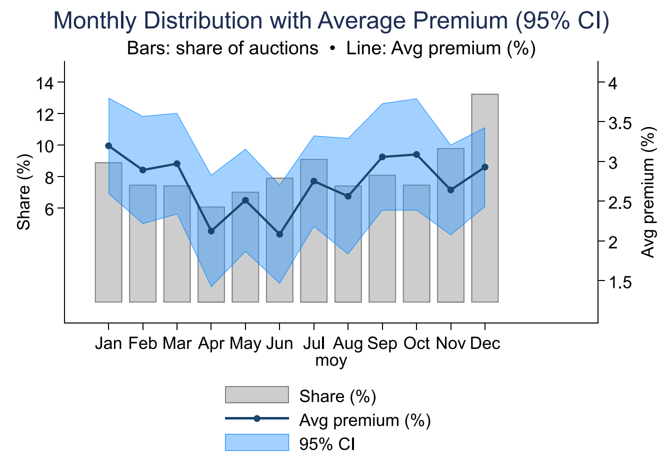



**Fig. 1.** Monthly distribution of auction shares and average premium.
*Note:* This figure shows the monthly distribution of land auctions (bars) along with the average premium paid in these auctions (line). The shaded area represents the 95% confidence interval (CI) around the average premium. The left $y$-axis indicates the share of auctions, while the right $y$-axis represents the average premium.

in most countries, including the United States, where almost all housing market transactions involve the joint sale of land and structures, Chinese local governments retain ownership of all urban land and lease land-use rights to developers. This institutional separation allows us to observe land prices and housing outcomes on distinct margins.

Second, land auctions are highly visible events. Large or expensive parcels routinely receive prominent media coverage, and they are closely watched by market participants. In Section 4.2.3, we show that search intensity for auction- and housing-related keywords on a major search engine spikes around auction dates, indicating that households actively track these auctions in real time. This salience makes it plausible that auction outcomes are perceived as informative signals about local housing market conditions.

Third, auction timing is largely administrative. We use the timing of land auctions to estimate their impact on nearby housing markets. A central concern is that governments might strategically schedule auctions in neighborhoods experiencing unobserved demand shocks. To mitigate this, we compare housing units within 3 km of an auctioned parcel to a local control group 3–5 km away. Our identifying assumption is that the exact month an auction closes is orthogonal to short-run changes in relative housing prices between these distance bands. This assumption is grounded in the institutional constraints described above: the transition from the annual plan to the auction block depends on technical factors, such as land assembly, utility installation, and bureaucratic approvals—rather than high-frequency fluctuations in local housing demand. Furthermore, auction cancellations are rare and typically result from a lack of bidders rather than government's strategic withdrawal. Consistent with this interpretation, Fig. 1 shows that auctions are distributed relatively evenly across months and that average premiums exhibit no systematic seasonal pattern, which is difficult to reconcile with deliberate clustering in high-demand months. We formally assess the plausibility of this timing assumption in Section 4.1.

### 2.2. Data and measurements

#### 2.2.1. Datasets

Our analysis builds on four unique datasets. These datasets provide rich information about China's land and housing market.

***The land auction dataset.*** We construct the land auction dataset from the Land Bureau of China (www.landchina.com). The data set includes about 56,000 land parcel transactions from 35 cities in China

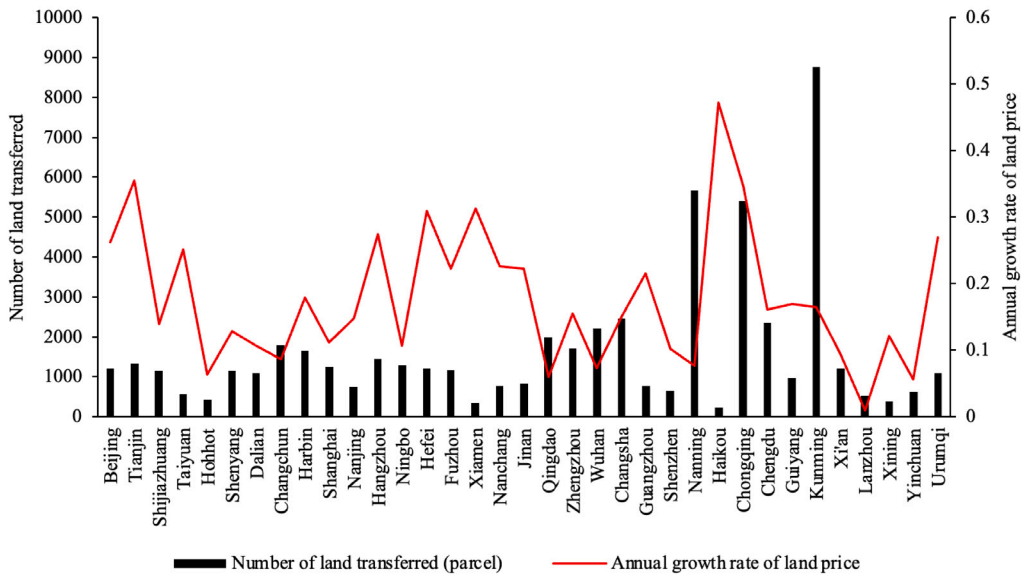



**Fig. 2.** Number of land parcels transferred and annual land price growth rate by city (2008–2017).
*Note*: This figure shows the number of residential land parcels transferred in 35 cities in China between 2008 and 2017, and the average annual growth rate of residential land prices.

between 2008 and 2017, recording detailed information about the transaction, including the area, auction type (two-stage auction or English auction), floor-to-area ratio, sale price, sale date and developer names who won the auction. We collect each parcel's longitude and latitude information from its name and address. Fig. 2 plots the number of land parcels supplied and the average growth rate of land prices between 2008 and 2017 separately for each city. The costs of land skyrocketed in this sampling period. The average annual growth rate of land prices in the 35 cities of China is 17%. This suggests that China experienced a prolonged housing boom market during this period, consistent with the patterns documented in Fang et al. (2016).

***The existing housing dataset.*** We obtained the housing transaction data set from Lianjia,[1] one of China's largest real estate brokerage firms in China. This dataset includes about 0.7 million housing transactions from 13 cities between 2010 and 2017. Each record provides the transaction price, transaction date, the name of the residential compound and detailed housing characteristics, such as the number of bedrooms. Unfortunately, the dataset does not record the detailed listing price adjustments and number of visits to the house. Therefore, we focus on the housing transaction prices instead of the listing price.

***The newly-built housing dataset.*** We use the sales information of newly-built housing projects from the China Index Academy (CIH), a real estate information and analytical service platform. This dataset covers about 0.6 million records of newly built houses from 35 major cities between 2008 and 2017. Each record includes the name and address of the housing project, developer names, and the monthly average transaction prices and volumes.

***House price expectation survey (HPES).*** We use the HPES, a survey designed to elicit households' expectations of future housing prices, to investigate the channels of land auctions on local housing markets. HPES was conducted quarterly during 2012–2013 in seven metropolitan cities (Beijing, Shanghai, Tianjin, Chengdu, Shenyang, Wuhan, and Xi'an) in China by the National Bureau of Statistics of China. Each quarter, households were interviewed with questions measuring their expectations of short-term and long-run housing prices.[2] Households also report detailed information about household demographic characteristics, with additional data on income and asset portfolio holdings. Notably, this survey employed a yearly tracking approach, interviewing the same respondents across four quarters each year. This allows us to observe how individuals change their short-term and long-run housing price expectations when observing the land auction outcomes.

The spatial relationships embedded in these data are central to our identification strategy, which we leverage in two ways. First, to estimate the effect of land-auction outcomes on nearby housing markets, we assign each housing unit to a pre-specified concentric ring around the auctioned parcel based on computed pairwise distances. This geographic partitioning is the foundation of our research design. Second, to analyze developers' bidding behavior, we identify whether a developer has existing projects near the auctioned land using the same geocoded data. Fig. 3 illustrates this spatial relationship for two parcels won by the same developer, and the comprehensive auction records allow us to apply this classification consistently.

### 2.2.2. Construction of the land auction premium

The core signal in our analysis is the land premium: how much faster the auctioned parcel's price grows relative to nearby housing prices within 3 km of the site. This premium summarizes developers' willingness to pay for land relative to the value of housing in the same micro-market, the cross-market cue that market participants observe and news coverage often highlights.

Construction proceeds in three steps. First, we measure the average monthly neighborhood housing price growth rates using a pseudo-repeat-sales method at the residential-community level (Guo et al., 2014). We calculate the average monthly house price growth over the preceding twelve months for each residential community within a 3 km radius of the auctioned parcel, then compute the average of these community-level growth rates. Second, land price growth is measured as the monthly growth rate from the price of the first parcel sold within the same 3 km radius in the preceding twelve months to the auction

[1] https://bj.lianjia.com/ershoufang/

[2] Respondents were asked, "How much do you think the house price in your city will increase in the next year (in percentage)?" and "How much do you think the house price in your city will increase in five years from now (in percentage)?". These two questions measure households' expectations of short-term and long-term housing prices.

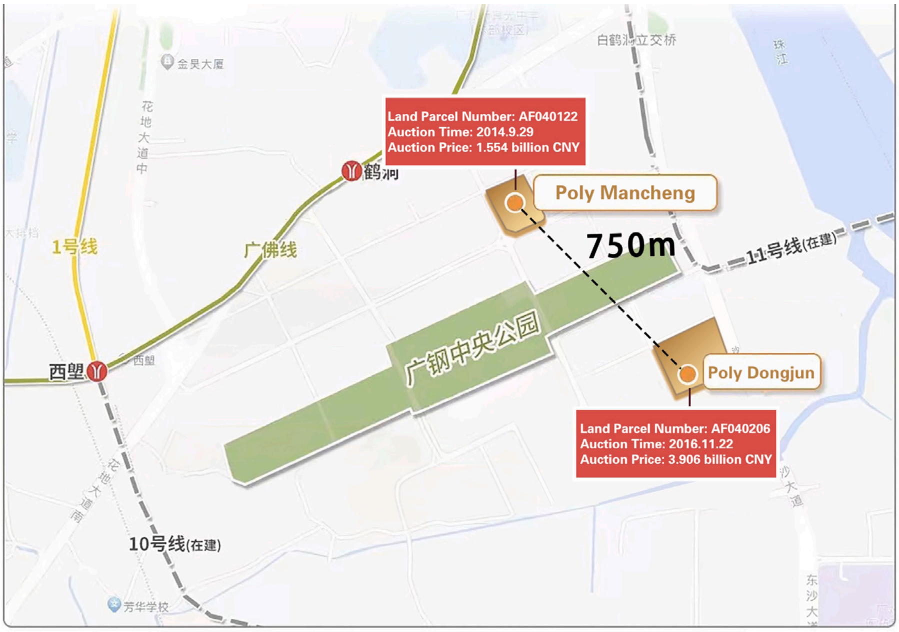



**Fig. 3.** Proximity and auction prices of two land parcels in Guangzhou acquired by the same developer.
*Note:* This figure shows the locations of two residential developments, Poly Mancheng and Poly Dongjun. Poly Mancheng (Land Parcel AF040122) was auctioned on September 29, 2014, for 1.554 billion CNY, while Poly Dongjun (Land Parcel AF040206), located 750 m away, was auctioned on November 22, 2016, for 3.906 billion CNY.

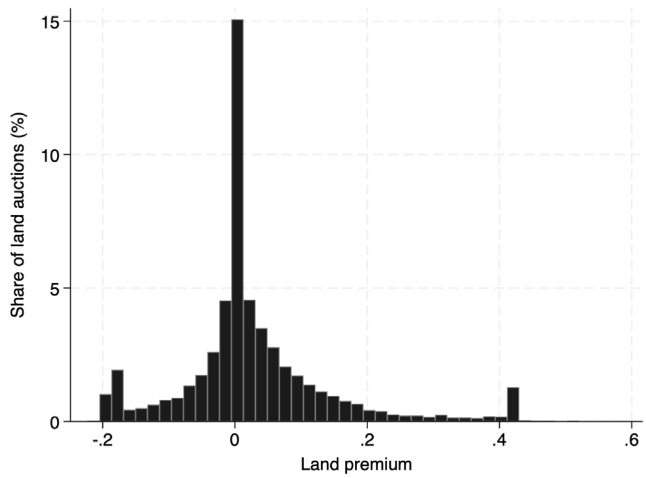



**Fig. 4.** Histogram of land premiums.
*Note:* This figure shows the histogrm of the land premium. The premium is defined as the differences between the average monthly growth rate of land and housing prices. To be specific, we calculate the average monthly growth rate of land prices by comparing the land price from each auction to historical prices of nearby lands, and the average monthly growth rate of newly-built housing prices. The premium is defined as the differences between these growth rates.

price of the focal parcel.[3] Third, we define the premium as land price growth minus housing price growth; a positive value indicates land appreciating faster than nearby housing.

Table 1 reports the summary statistics of the land premium. The mean premium is 0.028 with a standard deviation of 0.116, indicating significant variation. Fig. 4 plots the histogram of the premium index, revealing that a substantial fraction of land parcels are sold at a positive premium. Furthermore, Fig. 5 illustrates considerable cross-city heterogeneity. The average premium is highest in Urumqi, at 0.127, and lowest in Chengdu, at −0.003.

To ensure the robustness of our findings, we construct an alternative measure of the land premium. We refine our measure to enhance comparability by implementing a quality adjustment for land prices. For land, we estimate a hedonic regression controlling for parcel characteristics such as land area, floor-area ratio, land type, and transfer method, and then construct a quality-adjusted land price growth rate using the predicted residual values (see Appendix Table A1). For housing, the pseudo-repeat sales method already provides a quality-adjusted index. Our alternative premium is calculated from these adjusted series.[4]

## 3. Conceptual framework

We use a simple model to illustrate the mechanisms that developers attempt to use land auction bids to strategically influence the housing market and how this depends on home buyers' reactions to land auction results. The model features interconnected land and housing markets, where higher than expected land auction prices increase demand for nearby housing, and developers consider their auction bids taking into account their impacts on the housing market. Subsequently developers engage in oligopolistic price competition locally. We empirically test the model's predictions in Section 5.

Consider two land parcels and two property developers: land $a$ was already acquired by developer $i$ (the incumbent), and land $b$ is being auctioned where both developers $i$ and $j$ (the entrant) are interested. We assume that the cost of acquiring land $b$ is $c_0 + c_1 d_b$ which consists of two components: $c_0$ is the cost of acquiring the land at the prevailing market price, and $c_1 d_b$ is the extra cost that is increasing in the premium of the bid, where $d_b$ is the premium relative to the prevailing market price paid by the winning developer in acquiring land $b$ and $c_1$ is a

[3] A land auction is excluded from the analysis sample when the historical price data for calculating the growth rate of land or house price is unavailable.

[4] To mitigate the influence of noise in areas with thin markets, we also recalculate our main premium measure after excluding all auctioned parcels with fewer than three nearby land transactions in the preceding year.

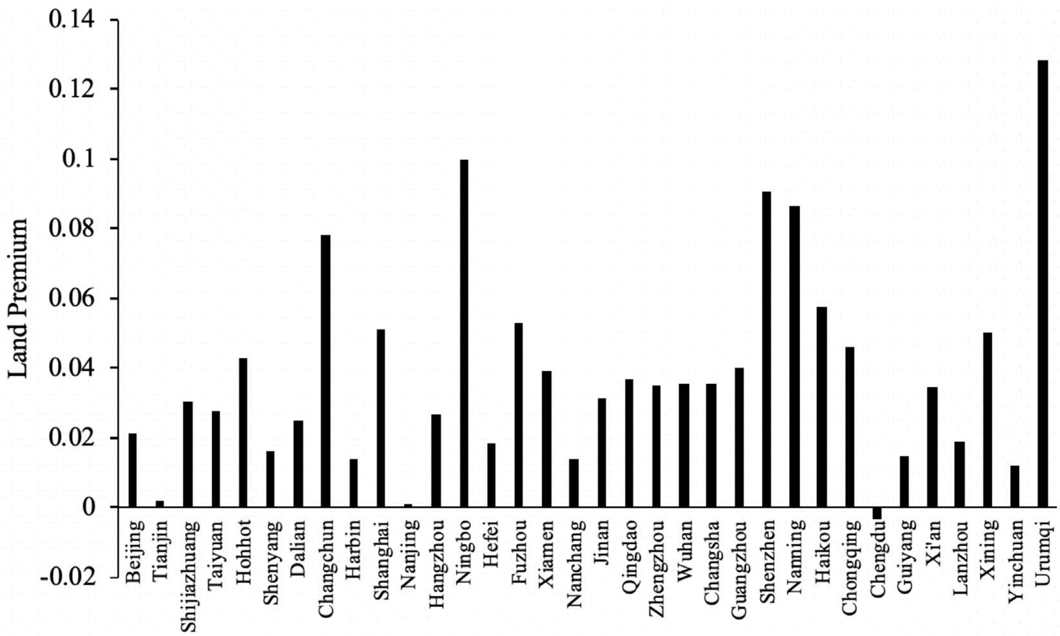



**Fig. 5.** Average land premium by city (2008–2017).
*Note:* This figure plots the average land premium separately for the 35 cities in China between 2008 and 2017. The premium is defined as the differences between the average monthly growth rate of land and housing prices. To be specific, we calculate the average monthly growth rate of land prices by comparing the land price from each auction to historical prices of nearby lands, and the average monthly growth rate of newly-built housing prices. The premium is defined as the differences between these growth rates.

**Table 1**
Summary statistics of land auctions, housing transactions, and household expectations.

| Variable | Obs | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| **Panel A. Land auction** | | | | | |
| Premium | 15,649 | 0.028 | 0.116 | −0.223 | 0.520 |
| Adjusted premium | 15,649 | 0.028 | 0.141 | −0.379 | 0.582 |
| Has nearby land stock | 15,649 | 0.094 | 0.292 | 0 | 1 |
| Land area (in hectares) | 15,648 | 4.665 | 5.285 | 0.001 | 114 |
| **Panel B. Existing housing sales** | | | | | |
| Housing prices (RMB per $m^2$) | 338,868 | 29,472 | 17,882 | 144 | 194,245 |
| Square meters | 338,868 | 86.667 | 83.469 | 5 | 14,085 |
| Number of bedrooms | 338,868 | 2.107 | 0.814 | 0 | 9 |
| Number of living rooms | 338,868 | 1.284 | 0.564 | 0 | 5 |
| Total number of floors in the building | 338,868 | 14.016 | 9.042 | 0 | 65 |
| **Panel C. Newly-built housing sales** | | | | | |
| Housing prices (RMB per $m^2$) | 403,795 | 11,211 | 9117 | 1013 | 420,705 |
| Transaction volumes | 379,568 | 6 | 9 | 0 | 165 |
| **Panel D. Households' expectation** | | | | | |
| Housing price expectation (12 months, %) | 10,553 | 2.729 | 8.109 | −50 | 30 |
| Housing price expectation (5 years, %) | 10,553 | 5.187 | 16.073 | −50 | 80 |

*Note*: This table reports the summary statistics of the main sample. The first row in Panel A reports the summary statistics of land premium, defined as the differences between the average monthly growth rate of land and housing prices. To be specific, we calculate the average monthly growth rate of land prices by comparing the land price from each auction to historical prices of nearby lands, and the average monthly growth rate of newly-built housing prices. The premium is defined as the differences between these growth rates. The second row in Panel A reports the summary statistics of the adjusted land premium discussed in Section 2.2.2. The third row in Panel A reports the summary statistics for whether the winning developer has other housing projects near the auctioned land parcel. Panel B and C report the summary statistics of transaction prices and other housing characteristics, distinguishing between existing and newly-built houses. Panel D provides data on households' expectations regarding changes in housing prices over the next 12 months and 5 years. This calculation is base on a subsample of households residing within five kilometers of a land auction during the sampling period.

fixed parameter. We assume that households' demand for housing in the local market of land parcels $a$ and $b$ follows a linear function:

$$h_a = h_a(p_a, p_b, d_b) = q_a + p_a\beta + p_b\lambda + d_b\alpha_a, \tag{1}$$

$$h_b = h_b(p_a, p_b, d_b) = q_b + p_a\lambda + p_b\beta + d_b\alpha_b, \tag{2}$$

where $p_a$ and $p_b$ are house prices in deviation from the prevailing market price. The housing demand equals $q_a$ and $q_b$ when there are no deviations in housing or land prices from the market prices, and we interpret $q_a$ and $q_b$ as measuring the fundamentals in housing demand. $\beta < 0$ is the own price effect on demand. The housing demand model is an oligopolistic competition model with heterogeneous products and linear price effects (Mobley, 2003).

In contrast to standard demand functions, we make the key assumption that housing demand increases with higher auction premium of nearby land, measured by positive $\alpha_a$ and $\alpha_b$. We do not directly model the underlying force driving this effect but provide a micro-foundation in Appendix A based on extrapolative beliefs. Intuitively, housing serves

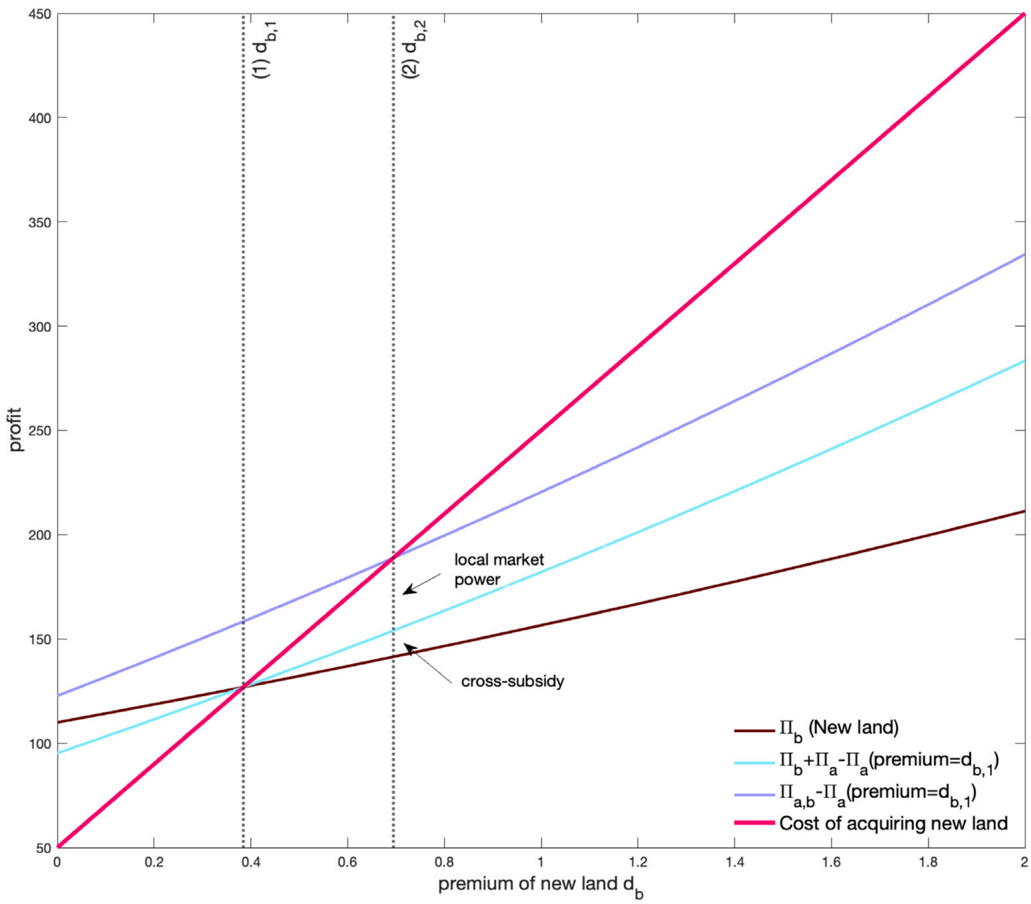



**Fig. 6.** Profits and costs of land auction bids.
*Note*: Parameter values: $q_a = q_b = 5$, $\beta = -0.1$, $\lambda = 0.05$, $\alpha_a = 0.8$, $\alpha_b = 1$, $c = 0.5$, $c_0 = 50$, $c_1 = 200$. The brown line ($\Pi_b$) is the profit of an entrant from land development $b$ assuming that the entrant and the incumbent engage in oligopolistic competition, which is increasing in the land premium $d_b$ because $\alpha_b > 0$. The purple line ($\Pi_{a,b} - \Pi_a(\text{premium} = d_{b,1})$) indicates the profit from an incumbent which owns both land development $a$ and $b$, relative to owning land $a$ only assuming that the incumbent will pay $d_{b,1}$ premium in acquiring land $b$. The light blue line indicates the additional profit for an incumbent but assuming that it is choosing $p_a$ and $p_b$ independently. The qualitative results regarding $d_{b,1}$ and $d_{b,2}$ still hold for other parameter values if an interior equilibrium exists. To see this, assuming that the interactions between the profit functions and the cost functions exist. Because when $\alpha_a > 0$ and $\lambda \neq 0$, $\Pi_{a,b} - \Pi_a(\text{premium} = d_{b,1}) - \Pi_b > \Pi_a + \Pi_b - \Pi_a(\text{premium} = d_{b,1}) - \Pi_b > \Pi_a - \Pi_a(\text{premium} = d_{b,1}) > 0$ when premium $> d_{b,1}$, the net monopoly profit is above the incumbent profit to the right of $d_{b,1}$, hence $d_{b,2} > d_{b,1}$. The light blue line is between the purple and the brown line to the right of $d_{b,1}$, because $\Pi_{a,b} > \Pi_a + \Pi_b$ and $\Pi_a - \Pi_a(\text{premium} = d_{b,1}) > 0$ when premium $> d_{b,1}$.

both as consumption good and financial asset. As a result, extrapolative households may believe that future housing price would increase after observing high land price and increase their housing demand. We investigate whether this assumption holds in our empirical analysis exploiting the exogeneity of the auction dates in China's land market.

**Assumption 1** (*Extrapolative Housing Demand*)**.** Assuming that $\alpha_a > 0$ and $\alpha_b > 0$, housing demand and house prices increase with nearby land premium.

Given how home buyers react to land premium, we now turn to the strategies of developers in land auctions. The profit functions of developers are

$$\pi_a = (p_a - c)h_a(p_a, p_b, d_b),$$
$$\pi_b = (p_b - c)h_b(p_a, p_b, d_b),$$

where $c$ is the marginal cost. Developers choose prices to maximize profits from property development.

If the new land $b$ is acquired by developer $i$ which already owns nearby land $a$, the developer acts like a monopoly in the local housing market and chooses $p_a$ and $p_b$ simultaneously to maximize profits:

$$\Pi_{a,b} = \arg\max_{p_a, p_b} \pi_a + \pi_b.$$

On the other hand, if the new land $b$ is acquired by developer $j$ (entrant) which does not own land nearby, developers $i$ and $j$ will act like a duopoly in the local housing market and engage in oligopolistic price competition. Developer $i$ chooses $p_a$ to maximize $\pi_a$ given $p_b$, and similarly developer $j$ chooses $p_b$ to maximize $\pi_b$ given $p_a$. In equilibrium, the prices satisfy

$$2\beta p_a^* + \lambda p_b^* = c\beta - d_b\alpha_a - q_a,$$
$$\lambda p_a^* + 2\beta p_b^* = c\beta - d_b\alpha_b - q_b.$$

The equilibrium profits under duopoly are denoted by $\Pi_a$ and $\Pi_b$ for developers $i$ and $j$, respectively. $\lambda$ is the cross price effect on demand. We make the following assumption on the value of $\lambda$.

**Assumption 2** (*Cross Price Effect*)**.** $\lambda \neq 0$.

With Assumption 2, monopoly profit ($\Pi_{a,b}$) is greater than duopoly profit ($\Pi_a$+$\Pi_b$). First, because the monopoly can implement the pricing strategies of duopolies, $\Pi_{a,b} \geq \Pi_a + \Pi_b$. Second, the inequality is strict because of demand spillovers ($\lambda \neq 0$), as a monopoly can internalize the effect of one unit's pricing on the demand of nearby units, thereby obtaining higher profits than duopolies.

Our model provides direct implication of developers' willingness to pay for the land parcel $b$. Fig. 6 compares the revenue and cost

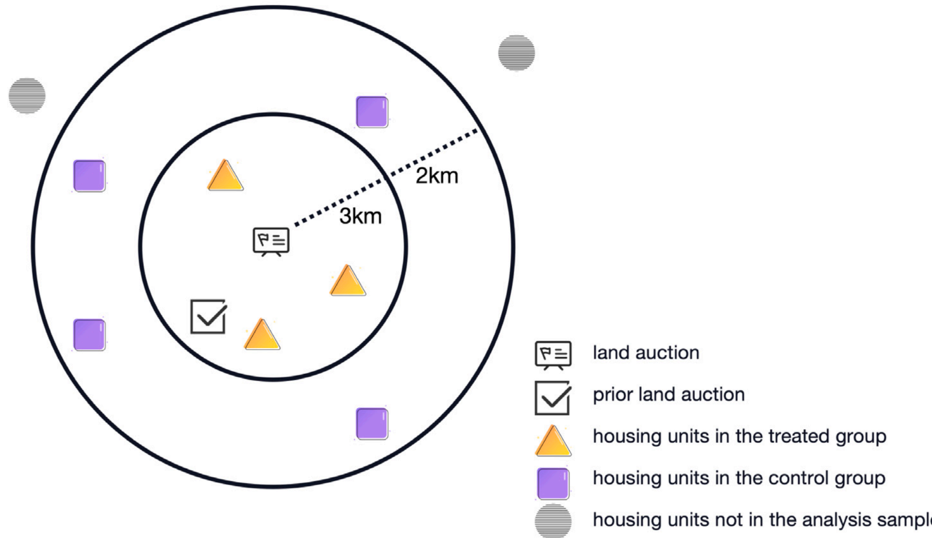



**Fig. 7.** Illustration of the empirical design.
*Note:* This figure illustrates the treated and control units for a land auction in the difference-in-differences empirical design. We compare changes in house prices before and after the land auction, between the treated and control units.

of developers $a$ and $b$, for a set of reasonable parameter values. The vertical lines (1) and (2) show the maximum premium that the entrant and that the incumbent developer $i$ are willing to pay, respectively. For the entrant firm, it is willing to increase the bids until the cost of acquiring the new land $b$ equals the profit from land $b$. For the incumbent developer $i$, its willingness to pay depends on the difference between the monopoly profit from owning $a$ and $b$ jointly and the duopoly profit from owning only $a$, net of the additional cost of acquiring $b$. It is clear that the incumbent developer $i$ bids higher than the entrant $j$. This heightened willingness to pay stems from two sources: the cross-subsidy effect, where the land premium hike increases demand for existing housing development, and increased profits attributable to a local market monopoly.

The predictions in Fig. 6 remain valid under a wider range of parameter values, provided that Assumptions 1 and 2 hold. We formalize these results in the following hypothesis:

**Hypothesis 1.** Assuming that Assumptions 1 and 2 hold, and an interior equilibrium exists, the developer who already owns land near the land being auctioned is willing to pay higher premium to acquire the land compared with developers which do not have nearby land holdings ($d_{b,2} > d_{b,1}$ in Fig. 6). The higher willingness to pay is due to two reasons: cross-subsidy where land premium from the new land auction increases demand for existing housing development (**Hypothesis 1A**) and higher profits for developers with greater local market power (**Hypothesis 1B**).

## 4. Local housing market responses to nearby land auctions

This section presents empirical evidence on how land auction outcomes impact the local housing market. Section 4.1 outlines our identification strategy. We present our main results in Section 4.2. We find that high auction premiums generate significant momentum in local housing prices and transaction volumes and provide evidence that this effect is driven by households revising their price expectations upward in response to auction signals.

### 4.1. Empirical strategy

In this subsection, we lay out our empirical strategy, econometric specification, and identification assumptions for estimating how nearby land auctions affect local housing prices.

Central to our model's prediction is Assumption 1, positing that households exhibit increased demand for housing when a nearby land parcel is sold at a high auction premium. However, straightforwardly comparing housing prices near land parcels with high auction prices against those further away does not suffice to validate Assumption 1. The reason is that houses near land parcels sold at high premiums are more likely to enjoy better amenities. We can control for some of the differences in attributes, but should be concerned that these homes also differ along unobserved attributes, such as neighborhood quality, expected infrastructure improvements, or local school reputation.

To address these concerns, we follow (Anenberg and Kung, 2014) to look at the changes in local housing prices in the few months surrounding a new nearby land auction. Fig. 7 depicts our identification strategy. For each auction, we link its land-price premium to housing transactions that occur within a 5 km radius.[5] Homes located within 3 km of an auction form the treatment group, while those 3–5 km away serve as controls.[6] The estimation window covers the six months preceding each auction and the eleven months that follow. Comparing the dynamics of transaction prices for homes just inside and just outside the 3 km radius around the auction date yields an event-study style Difference-in-Differences design.

Specifically, the empirical model is as follows:

$$y_{ikjt} = \alpha_0 + \sum_{n\neq -1} \alpha_1^n \widetilde{\text{premium}}_k \times \text{land}_{it}^n + \sum_{n\neq -1} \alpha_2^n \text{land}_{it}^n + \rho_k + \omega_{jt} + \tau_t + \beta_1 Z_{kt} + \beta_2 X_i + \varepsilon_{ikjt}, \tag{3}$$

where $y_{ikjt}$ is the log price of housing unit $i$ in residential compound $k$ in city $j$ and month $t$. The $\widetilde{premium}_k$ is the centered index formally defined in Section 2.2, measuring the relative growth rate of the land price near

[5] The premium is calculated as detailed in Section 2.2. Because the computation requires growth rates for both land prices and neighboring house prices, we exclude auctions that lack (i) a prior land-auction record within 3 km or (ii) nearby housing-price data. When several qualifying auctions map to the same neighborhood, we keep only the earliest observation with an available premium.

[6] We choose the bandwidth of the concentric ring as 3 km by investigating the spatial attenuation pattern of the auctions' impacts on the housing market. Appendix A discusses this procedure in detail.

the residential compound $k$.[7] $land^n_{it}$ is the dummy variable indicating whether the housing unit $i$ is located within a 3-km of any land auction in $n$ months relative to the month $t$ of the auction. We omit $n = -1$ so that month $-1$ serves as the reference period. $\rho_k$ is the residential compound fixed effect. $\omega_{jt}$ measures the city-by-year fixed effects and $\tau_t$ is the year-month fixed effects. $Z_{kt}$ controls for compound level characteristics such as the log of the total amount of land auctioned within 3-km of residential compound $k$. $X_i$ controls various characteristics of the housing unit $i$. These characteristics include the square feet, direction, number of bedrooms, the number of floors of the house, and the total number of floors of the building. Together, these controls and fixed effects remove time-invariant neighborhood differences and city specific macro trends, allowing us to isolate within-neighborhood, within-period variation linked to nearby auction outcomes.

The parameters of interest are $\alpha^n_1$, measuring the change in the housing price located within 3 km of the land auction in months relative to the auction date, compared to the price of those located between 3 and 5 km. As assumed in our conceptual framework, we should observe positive $\alpha^n_1$ when $n \geq 0$. By contrast, the pre-auction coefficients for $n < 0$ should be close to zero if treated and control neighborhoods follow parallel trends before the auction. The coefficients $\alpha^n_2$ on the event-time dummies $\text{land}^n_{it}$ capture any average price differences between homes within 3 km and those 3–5 km away at each event month. Allowing for these non-interacted terms means that prices in the inner and outer rings can follow different baseline paths around the auction date.

Our identification relies on a parallel-trends assumption: absent land auctions, housing prices for units near and far from auctioned parcels would have evolved similarly over event time, conditional on compound fixed effects, city-by-year fixed effects, calendar time effects, and unit characteristics. This assumption is plausible in our setting because, as discussed in Section 2.1 on institutional background, the exact timing of land auctions is driven primarily by administrative and procedural considerations, such as the approval process of annual land-supply plans and internal scheduling constraints, rather than short-run neighborhood-specific shocks to housing demand. In this sense, auction dates are as good as randomly assigned with respect to local housing price shocks.

The construction of the premium index further strengthens identification for $\alpha^n_1$. The premium index is defined as the difference between land price growth and nearby housing price growth within 3 km of the auction site, constructed using only pre-auction information. Since the premium is constructed from pre-auction data, it is predetermined with respect to post-auction housing prices, and our rich set of fixed effects absorbs city-level and neighborhood-level trends. Under the assumption that, conditional on these controls, the premium is not systematically correlated with unobserved shocks to nearby existing-house prices, the $\alpha^n_1$ can be interpreted as capturing how a stronger land bidding signal translates into larger revisions in local housing prices.

To evaluate the credibility of this identification strategy, we examine whether land auction timing is plausibly exogenous. We exploit variation in housing market conditions at the district and monthly levels to test whether local demand shocks drive land auction timing. We regress subsequent auction activity, including the number of auctions and the total land area transacted within a district, on lagged local housing price levels and housing price growth rate. As reported in Table A3, neither lagged housing prices nor recent price growth systematically predict future auction activity.[8] These results suggest that short-run fluctuations in local housing market conditions do not drive the timing or intensity of land auctions.

Together, the institutional structure and these diagnostics indicate that short-run demand dynamics do not materially drive auction scheduling, mitigating concerns that our results are artifacts of endogenous timing. Finally, in our main event-study specifications we explicitly examine the pre-auction coefficients for both $\alpha^n_1$ and $\alpha^n_2$. The absence of systematic pre-trends in these coefficients provides additional support for the parallel-trends assumption underlying our identification.

### 4.2. Empirical results

In this subsection, we first present two-way fixed effects (TWFE) estimates of how nearby land auction premiums affect local housing prices and transaction volumes. We then show that these results are robust to using a quality-adjusted land premium and to applying an interaction-weighted DID estimator that accounts for treatment-effect heterogeneity under staggered adoption. Finally, we complement these reduced-form results with survey evidence on households' housing price expectations, which allows us to probe the underlying belief-updating mechanism. The data and code for replicating the results in this paper are available in Liu et al. (2026).

#### 4.2.1. Main estimates

Fig. 8 plots pre-treatment and post-treatment trends separately for existing and newly-built housing transactions. First, before the land auction, the housing prices near the auctioned land are similar to those further away, and the difference is small and statistically insignificant. The pre-treatment trend validates our use of the DID strategy. Second, during the precise month of an auction, there is a notable surge in housing transaction prices. In the month of the auction, a one standard deviation increase in the land premium leads to a 1.7 percent change in the transaction price of houses located within 3-km of the land parcel, relative to those located further away.[9] This effect demonstrates considerable momentum, persisting for three months post-auction before a subsequent adjustment downwards. The pattern of an initial positive shock followed by a reversal suggests that the price impact of land premiums is transitory rather than permanent. The right panel illustrates similar results for newly-built houses. Pre-treatment differences are small and statistically insignificant. Newly-built housing prices rise significantly following high-premium auctions and continue to increase post-auction.

Splitting our sample into upturn and downturn periods, depicted in Fig. 9, suggests that the effects differ across market cycles. Point estimates are larger and statistically significant during upturn periods, while the estimates are smaller and generally insignificant during downturns.[10]

[7] Let $\widetilde{\text{premium}}_k = \text{premium}_k - E\left(\text{premium}_k\right)$. $\widetilde{\text{premium}}_k$ is used in the interactions. With this variable centered, the coefficient of $\text{land}_{it}$ represents the average effect of nearby land auctions, with and without the interaction term. Note that $\widetilde{\text{premium}}_k$ is absorbed into the fixed effects and hence not separately reported.

[8] Because the dependent variables contain many zeros (district-months with no auction), we estimate the model using Poisson pseudo-maximum likelihood (Cohn et al., 2022).

[9] The complete set of regressions results used to plot Figures are reported in Appendix D. The standard deviation for the land premium is 0.12. The 1.7 percent price increase is calculated by multiplying the standard deviation of the land premium with the corresponding coefficient, 0.1379, reported in Table A5.

[10] The housing upturn period is defined when a city's housing price growth rate is higher than the city's GDP growth rate. And the housing downturn period is defined as the opposite. Fig. 9 only reports the results using the newly-built housing dataset. We do not include the results using the existing houses dataset due to its limited sample coverage of a representative set of cities. More than 50% of the observations in the existing houses dataset are transactions in Beijing. Therefore, when separating the sample into housing upturns and downturns, the effect in different housing cycles in Beijing

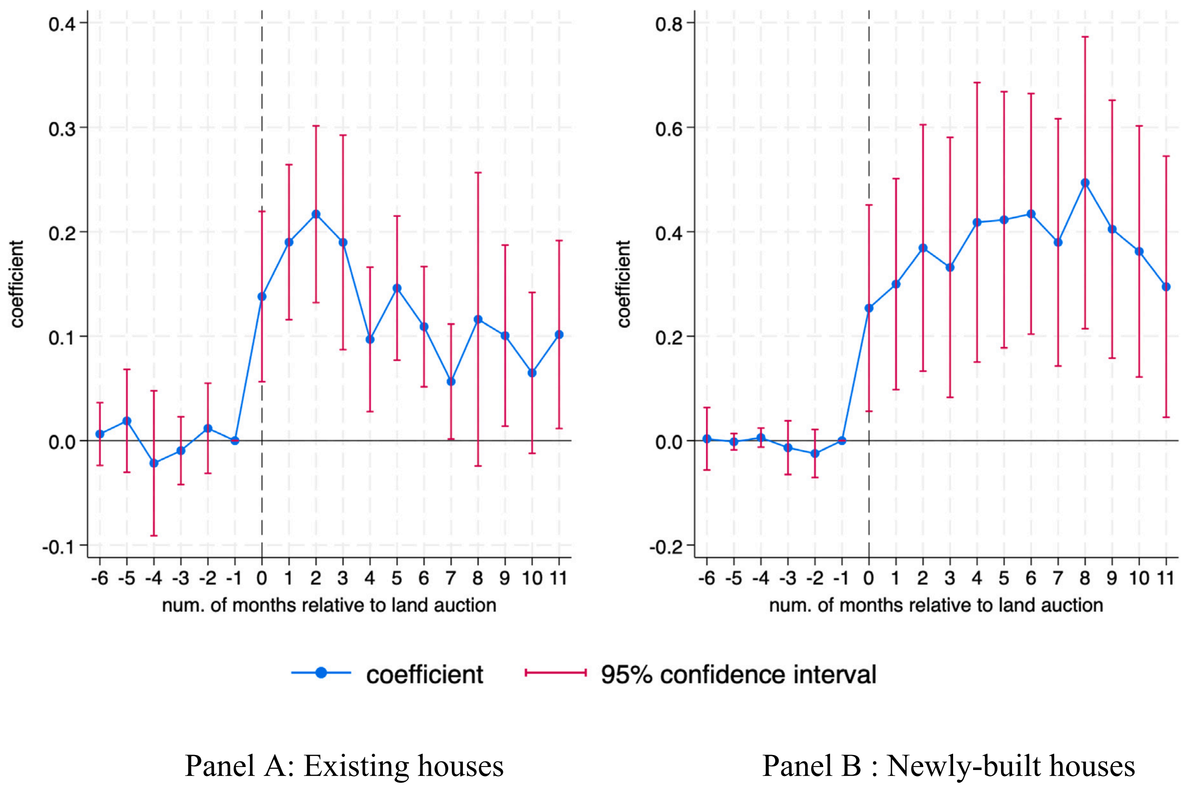



**Fig. 8.** Dynamic effects of land auction premium on local housing prices for existing and newly-built houses.
*Note*: This figure reports the value of $\alpha_1^n$ from estimating Eq. (3) using the TWFE estimator separately for existing and newly-built houses. The $y$-axis is the coefficient value and the $x$-axis is the number of months relative to the month where the land auction is held. For existing houses, we control the residential compound, year-month, and city by year fixed effects, and various housing characteristics including square feet, direction, number of bedrooms, the number of floors of the house, and the total number of floors of the building. For newly-built houses, we control the residential, year-month, and city by year fixed effects.

Theoretical frameworks focusing on extrapolative households also predict increased transaction volumes following higher returns (Barberis et al., 2018; Liao et al., 2022; DeFusco et al., 2022). To test this prediction, we estimate Eq. (3) using the monthly transactions of newly-built homes as the dependent variable.[11] Fig. 10 presents the dynamic effects of land auction outcomes on nearby housing transaction volumes. In the six months prior to the auction, the coefficients are small and statistically indistinguishable from zero, confirming the absence of pre-trends or anticipatory effects. Coinciding with the auction month, we observe an immediate and statistically significant surge in transaction volume. Notably, this effect persists throughout the post-auction window

#### 4.2.2. Robustness

***Alternative measure of land premium.*** To ensure that our main findings are not driven by unobserved changes in the quality or composition of the transacted land parcels, we re-estimate our models using the quality-adjusted land premium described in Section 2.2.2. This measure removes the land price growth series of the influence of parcel characteristics, providing a cleaner signal of price appreciation that is directly comparable to the quality-adjusted housing price index.

We begin by re-estimating our baseline model, replacing the main land premium variable with this alternative measure. Panel A and B in Figure A1 plots the pre- and post-treatment trends using the quality-adjusted premium. The qualitative pattern is nearly identical to that in our main analysis: a sharp price surge coinciding with the auction month, persisting for approximately three months before a subsequent attenuation. This confirms that the transitory nature of the price effect is not an artifact of changing land quality.[12]

Furthermore, we note that the asymmetry across the housing cycle is less pronounced when using the alternative measure of land premium, as shown in Panel C and D in Figure A1. Finally, we reassess the impact on transaction volume. Panel E in Figure A1 illustrates that higher quality-adjusted land premiums are also associated with a significant increase in purchases of newly-built houses post-auction.

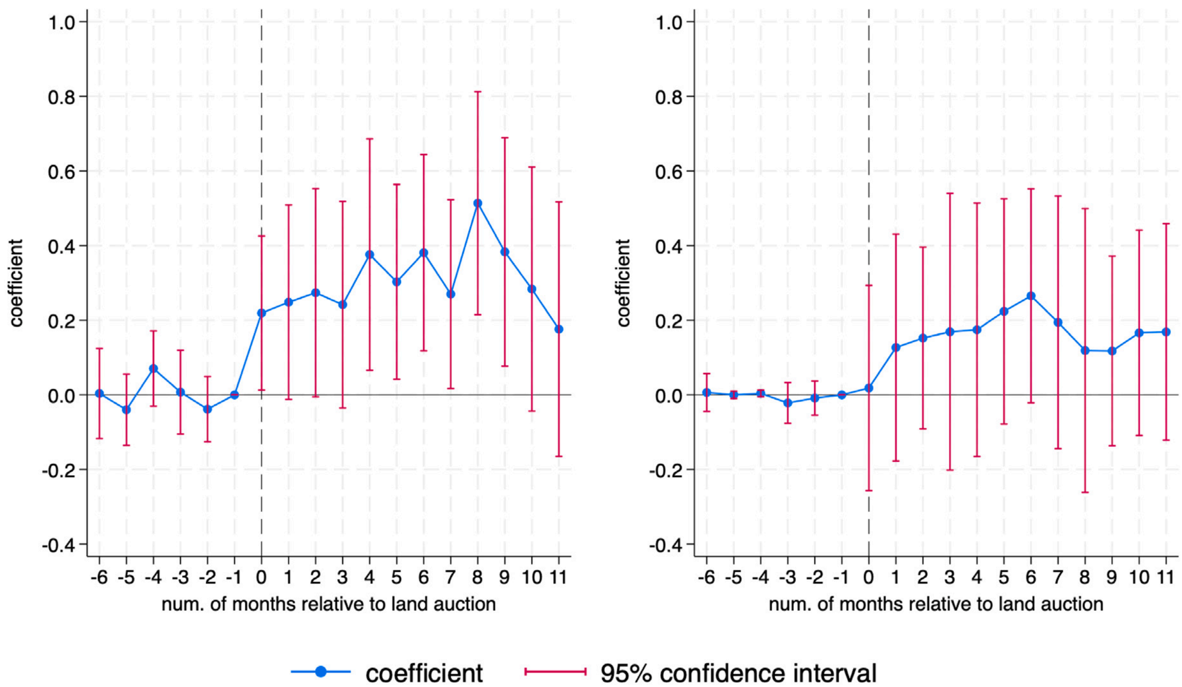
In summary, the robustness of our results to this quality-adjusted measure increases confidence that we are capturing the causal effect of the land premium signal itself, rather than spurious correlations driven by parcel heterogeneity.

***Heterogeneous treatment effects.*** Recent studies have highlighted the potential bias in traditional DID estimates due to heterogeneous treatment effects and staggered adoption. This issue arises particularly when treatments are introduced in a staggered fashion across a lengthy panel data set (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; De Chaisemartin and D'Haultfoeuille, 2020). Our data is an unbalanced panel of housing transactions with variation in the timing of the matched land auctions. Although a housing unit in the treatment (0–3 km) or control group (3–5 km) of one land auction is never matched to another land auction, the staggered timing of land auctions implies that, in a two-way fixed effects regression model, housing units with no changes in their treatment status, including

dominates other cities. The results for the existing houses are available upon request.

[11] The existing houses data only include housing transactions brokered through the real estate agency of Lianjia. Since Lianjia's market share varies in different cities, we do not use this source to calculate the land auctions' impacts on transaction volumes.

[12] The detailed estimation results are presented in Appendix Table A6.




Panel A: Upturn period

Panel B: Downturn period

**Fig. 9.** The effect of land premiums on the local housing prices in upturn and downturn periods.
*Note*: This figure reports the value of $\alpha_1^n$ from estimating Eq. (3) using the prices of newly-built houses in housing upturn and downturn periods separately as the dependent variable. We estimate Eq. (3) using the TWFE estimator. The housing upturn period is defined when a city's housing price growth rate is higher than the city's GDP growth rate, and the housing downturn period is defined as the opposite. The $y$-axis is the coefficient value and the $x$-axis is the number of months relative to the month where the land auction is held. We control the residential compound, year-month, and city by year fixed effects.

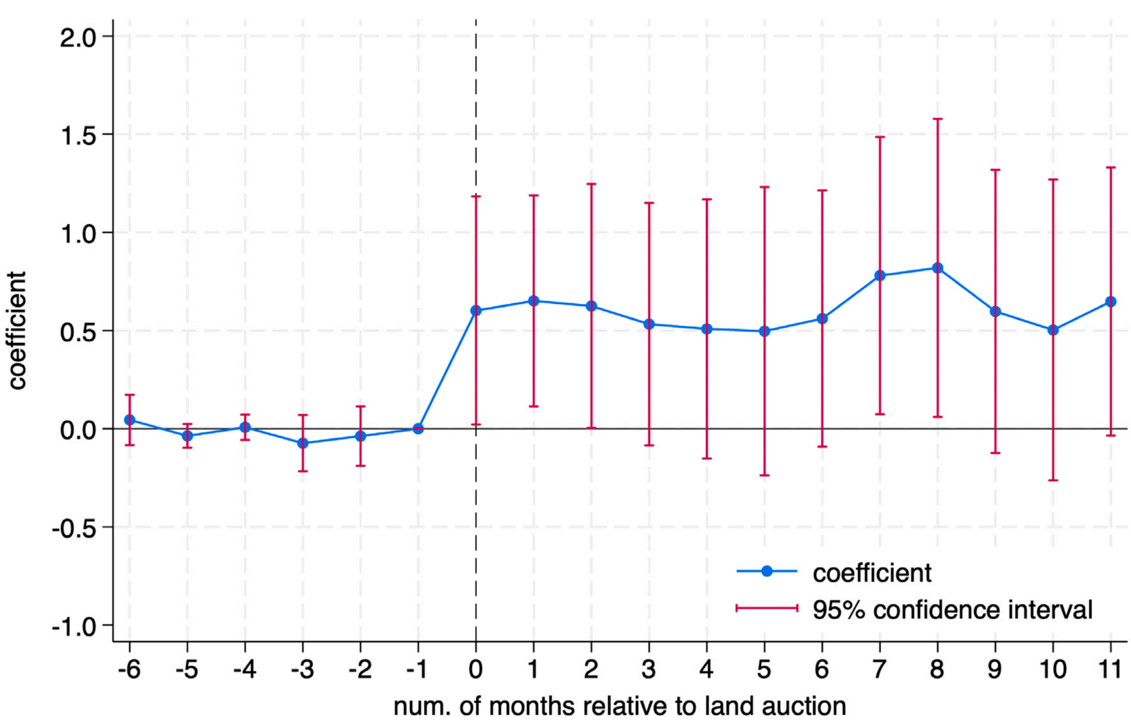



**Fig. 10.** Dynamic effects of land auction premiums on transaction volumes of newly-built houses.
*Note*: This figure reports the value of $\alpha_1^n$ from estimating Eq. (3) using the number of transactions of newly-built housings as the dependent variable. We estimate Eq. (3) using the TWFE estimator. The $y$-axis is the coefficient value and the $x$-axis is the number of months relative to the month where the land auction is held. We control the residential compound, year-month, and city by year fixed effects.

those already treated, will be used as control groups for housing units near subsequent land auctions. If the effects from land auctions vary over the years, standard two-way fixed effects regressions may yield biased estimates by using earlier-treated units as controls. To address these concerns, we complement our baseline OLS estimates with the heterogeneity-robust DID estimator proposed by De Chaisemartin and D'Haultfœuille (2024), which is designed to recover convex averages of cohort- and event-time-specific treatment effects under staggered adoption. We report results from the heterogeneity-robust DID estimator of De Chaisemartin and D'Haultfœuille (2024) in Appendix Figure A2. The results are consistent with the findings from the TWFE estimates.[13]

The De Chaisemartin and D'Haultfœuille (2024) estimates are directionally consistent with the TWFE baseline and always statistically significant; though their magnitudes are smaller, with TWFE coefficients roughly four to five times larger for newly-built houses. This difference is expected given that the two estimators target different weighted averages of cohort-by-period treatment effects (De Chaisemartin and D'Haultfoeuille, 2020; Callaway et al., 2024). Specifically, TWFE assigns larger weights to cohorts with higher residualized treatment values. In our setting, these correspond to boom-period auctions which generate higher premiums. Land premiums from boom-period auctions also produce stronger housing-price responses (Appendix Figure A2; Appendix Table A7, columns 3–4). TWFE therefore upweights precisely these cohorts with high treatment effects, while the De Chaisemartin and D'Haultfœuille (2024) estimator recovers a convex average weighted by cohort sample shares. The magnitude gap therefore reflects the heterogeneity in treatment effects across housing market cycles documented above.

#### *4.2.3. Mechanism*

We interpret the concurrent rise in prices and transaction volumes as evidence that households update their beliefs about local housing prices after observing high land premiums. Because developers are often viewed as informed agents with superior forecasting ability, land auctions provide a salient and credible signal about fundamentals in a market with substantial information frictions (Yung and Zender, 2010). When auctions clear at high premiums, households infer stronger future price growth and revise upward their expectations about local housing prices.

This belief-updating mechanism operates asymmetrically over the housing cycle. During upturns, high-premium auctions reinforce already optimistic beliefs and translate into sizable increases in transaction prices. By contrast, in downturns the same signals are insufficient to overturn prevailing pessimism, so price responses are muted. This pattern is consistent with Chan et al. (2016) and Shen et al. (2022), who show that homeowners adjust their subjective valuations more strongly when market conditions are favorable than when they are adverse.

Our findings also relate to theoretical models in which trading volume is generated by belief dispersion or realization utility (Barberis et al., 2018; Liao et al., 2022). Most of these frameworks, however, are developed for highly liquid financial markets. In contrast, housing is illiquid and subject to substantial transaction costs. In this environment, the volume spikes we document are more in line with the short-horizon trading motives emphasized by DeFusco et al. (2022): salient news about local fundamentals triggers a temporary reallocation of housing assets among households with heterogeneous beliefs and planning horizons. The high visibility of land auctions amplifies this response by drawing attention to the local market and coordinating expectations.

We next assess this mechanism through two tests that speak directly to attention and expectation updating. First, we examine whether auction outcomes are sufficiently salient to be observed and processed by households in real time. Second, we investigate whether households exposed to high-premium auctions indeed revise their expectations upward. Together, these tests allow us to evaluate whether attention and expectation updating operate as key mechanisms underlying our main results.

***Auction salience.*** Our interpretation of land auctions' impact on housing-price dynamics rests on the simple premise: auctions transmit information only if households actually see them. Land auctions are public, often videotaped, and widely covered by the press. But do households actually notice them in real time, and does that attention spike when record prices are set?

Case studies offer a compelling first look. For instance, a land auction in Guangzhou on November 16, 2020 garnered substantial attention. During this event, a land parcel (AT1003036) was sold for 4.8 billion yuan, setting a new record with a value exceeding fifty thousand yuan per square meter.[14] Following this record-breaking sale, city-level search activity on Baidu Index spiked and listing prices for nearby homes rose sharply. Panel A of Figure A3 maps the distance between the auctioned parcel and surrounding residential properties; Panel B displays a representative listing from Lianjia, a leading real estate brokerage in China. On the day the record price was set, listing prices increased by 8.5%, consistent with auctions attracting immediate local attention and potentially affecting housing prices.

To move from anecdote to systematic evidence, we test whether auctions trigger immediate, city-level spikes in search activity on Baidu Index, a high-frequency proxy for public attention. Baidu Index, the Chinese counterpart to Google Trends, provides high-frequency search volumes for specific keywords at the city–day level. This source has been used in economics to proxy public attention and information acquisition (Qin and Zhu, 2018; Alm et al., 2022). Its key virtue for our context is timeliness: search volumes move as soon as public attention shifts, allowing us to test whether auctions trigger immediate information demand among local households.

We construct a city-day panel by compiling daily Baidu Index series for housing and land-auction keywords and merging them with auction dates. Our empirical strategy employs a fixed-effects regression model:

$$\log(y_{it}) = \beta_1 \cdot \text{premium}_{it} + \beta_2 \cdot \text{Auction day}_{it} + \theta_i + \delta_t + \epsilon_{it} \tag{4}$$

where $\log(y_{it})$ is the logarithm of the Baidu Index for housing price or land auction keywords in city $i$ on day $t$, $\text{premium}_{it}$ is the average premium for land parcels sold in city $i$ on day $t$, $\text{Auction day}_{it}$ is an indicator equal to one if any land auction takes place in city $i$ on day $t$, $\theta_i$ denotes city fixed effects, and $\delta_t$ denotes time fixed effects whose granularity varies across specifications. Including the auction-day indicator separates the extensive margin (whether any auction occurs) from the intensive margin (how high the premium is).

Table 2 reports the results. All three columns include both the premium and the auction-day indicator on the full sample; columns differ in the stringency of time fixed effects. The premium coefficient is large and statistically significant across all specifications. Under the most demanding specification with city and date fixed effects (Column 3), a one-unit increase in the land premium raises search intensity by

[13] In our econometric specification, we interact the land auction dummy with a continuous treatment (land premium index). As Callaway et al. (2024) highlights, continuous treatments require stronger identification assumptions than binary treatments in Difference-in-Differences analyses. To address these concerns, we classify land auctions into high-premium groups if their premium exceeds zero and estimate a DID model with the binary treatment. The results are reported in column 5 of Appendix Table A5. We also experiment with various threshold values and find our results to be robust across alternative classifications.

[14] Source: Guangzhou Public Resources Trading Center. http://www.gzggzy.cn.

**Table 2**
Impact of land premiums on search intensity for housing-related keywords.

| | log(Search intensity) | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Premium | 0.4102 | 0.3587 | 0.3704 |
| | (0.1675)** | (0.1532)** | (0.1603)** |
| | [0.1116]*** | [0.1025]*** | [0.1054]*** |
| Auction day | 0.0433 | 0.0388 | 0.0511 |
| | (0.0212)** | (0.0208)* | (0.0239)** |
| | [0.0103]*** | [0.0097]*** | [0.0104]*** |
| City FE | Yes | Yes | Yes |
| Year FE | Yes | No | No |
| Year-month FE | No | Yes | No |
| Date FE | No | No | Yes |
| Observations | 89,495 | 89,495 | 89,495 |
| $R^2$ | 0.8098 | 0.8320 | 0.8351 |

Notes: The dependent variable measures Baidu search intensity for housing-related keywords at the city–date level. All columns include both the land premium and an auction-day indicator equal to one if any land auction takes place in the city on that day. Column (1) includes city and year fixed effects. Column (2) replaces year fixed effects with year-month fixed effects. Column (3) replaces year-month fixed effects with date fixed effects, the most demanding specification. Standard errors in parentheses are clustered at the city–year level; standard errors in brackets are heteroskedasticity-robust. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

approximately 45%.[15] The auction-day indicator carries a small but significant positive coefficient, indicating that auctions modestly raise search activity even apart from the premium; however, the premium effect is roughly an order of magnitude larger, confirming that it is the intensity of the price signal, not the mere occurrence of an auction, that drives the attention response.

To probe whether these effects could arise from spurious city-specific time patterns, we implement a randomization-inference exercise: holding auction dates fixed, we randomly reassign premium values across auction days within each city's calendar and re-estimate the Column (3) specification 500 times. Figure A4 shows that the observed coefficient lies in the extreme right tail of the placebo distribution, indicating that the measured effect is highly unlikely to be due to chance.

Taken together, our findings support the premise that land auctions are salient and immediately noticed by local households. The strong same-day response in Baidu searches for both housing and land-auction terms provides precisely the broad, real-time attention metric, strengthening our mechanism that auctions quickly transmit information into household behavior.

***Survey evidence on belief updating.*** To directly test whether households revise their expectations after observing nearby high-premium auctions, we turn to survey data from the HPES. We link households to land auctions using their home addresses and define "treated" households as those experiencing at least one auction within 3 km of their home in the survey quarter. Approximately 27% of respondents fall into this group and, conditional on treatment, we observe them on average 1.93 quarters after the auction. The control group consists of households living between 3 and 5 km from an auction site, mirroring the spatial comparison used in our main analysis.

The econometric specification is:

$$y_{it} = \beta_0 + \beta_1 \widetilde{premium}_i * land_{it} + \beta_2 land_{it} + \sigma_i + \theta_t + \omega_{jt} + \epsilon_{it}, \quad (5)$$

where the dependent variable is individual $i$'s expectation of future housing price growth rate measured in percentage terms in quarter $t$. The $\widetilde{premium}_i$ is the centered average land premium within 3-km of individual $i$'s home address. $land_{it}$ is the dummy variable indicating whether individual $i$'s home is located within 3-km of any land auction in quarter $t$. $\sigma_i$ and $\theta_t$ are individual and quarter-by-year fixed effects. $\omega_{jt}$ denotes the city-by-quarter fixed effects. $\beta_1$ is the parameter of interest and measures changes in households' expectations of future housing prices after a land parcel auction, relative to those who live further away. Standard errors are clustered at the household level.

Table 3 presents the estimation results. Columns (1) and (2) report the impact of land auctions on households' expectations of housing prices over the next 12 months. In column (1), we directly control for the city-level house price growth rate relative to the previous quarter. In column (2), we use city-by-quarter fixed effects to control for differential housing price trends across cities. We find that households exposed to land auctions with higher premiums have more optimistic short-term expectations about housing prices. Columns (3) and (4) replicate the analysis using 5-year housing price expectations, suggesting similar effects. Taken together with the robust price and volume responses documented above, these results support the view that high land premiums operate primarily by temporarily shifting households' beliefs about near-term housing price growth.

## 5. Developers' strategies in land auctions

Section 4 documents robust evidence of households' cross-market extrapolations. Our stylized model in Section 3 demonstrates that developers holding nearby land parcels can strategically bid to inflate land premiums, anticipating a corresponding rise in nearby housing prices that increases the value of their existing stock. In this section, we empirically test the core prediction from our conceptual framework: developers with nearby land stock pay high premium for land parcels. We employ a series of robustness tests and exploit a policy change in housing market to argue that the higher premiums arise from developers' strategic incentives.

### 5.1. Developers' strategies

We estimate the following baseline specification that links developers' land premiums to their existing local land holdings:

$$y_{ljt} = \lambda_0 + \lambda_1 Z_l + \lambda_2 X_l + w_{jt} + \epsilon_{ljt}, \quad (6)$$

where $y_{ljt}$ is the land premium of parcel $l$ in city $j$ and year $t$, defined in Section 2.2, which measures the growth rate of the land price relative to local house prices. $Z_l$ denotes whether the winning developer of land parcel $l$ holds other land within a 3 km radius. $X_l$ includes other characteristics of land $l$ such as the total land area, and $w_{jt}$ measures the city-by-year fixed effects. The parameter of interest is $\lambda_1$. According to Hypothesis 1 in our conceptual framework, we expect a positive $\lambda_1$, indicating that developers with nearby land stock bid more aggressively. We report both heterogeneous robust standard errors, and two-way clustered standard errors at the developer and city levels.

Table 4 presents the estimates from Eq. (6). Columns (1)–(3) show that developers who already hold land within a 3–km radius consistently bid more aggressively: in our preferred specification, controlling for city–by–year variation, land–use category, and auction method, these developers pay, on average, a 3.9 percent higher premium.[16]

[15] Because the dependent variable is in logs, the coefficient $\beta_1$ is a semi-elasticity. The implied percent change in $y_{it}$ from a one-unit increase in premium$_{it}$ is $100 \times (\exp(\beta_1) - 1)$. Thus, with $\beta_1 = 0.370$, $100 \times (\exp(0.370) - 1) \approx$ 45%.

[16] The land type refers to the designated use of the land, including middle and low-priced, medium and small-sized ordinary residential land, ordinary residential land, public rental housing land, other residential land, other ordinary residential land, urban residential land, affordable rental housing land, economically affordable housing land, and high-end residential land. The land transfer type includes the methods used to auction land discussed in Section 2.1.

**Table 3**
Impact of land premium on households' housing price expectations.

| | Housing price expectation (12 months) | | Housing price expectation (5 years) | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| $\widetilde{\text{Premium}}$ × Land | 3.7805 | 3.8357 | 4.8227 | 5.0627 |
| | (0.7156)*** | (0.7421)*** | (1.3914)*** | (1.4553)*** |
| | [0.7072]*** | [0.7353]*** | [1.4375]*** | [1.4960]*** |
| Land | −0.2735 | −0.5184 | −0.8875 | −0.9550 |
| | (0.4531) | (0.4774) | (0.8800) | (0.9063) |
| | [0.4545] | [0.4783] | [0.9045] | [0.9297] |
| log(Land area) | 0.0531 | 0.2073 | 0.1193 | 0.1321 |
| | (0.1403) | (0.1471) | (0.2742) | (0.2930) |
| | [0.1403] | [0.1493] | [0.2740] | [0.2910] |
| Δ log(Housing price) | 5.7139 | | 19.7967 | |
| | (3.8935) | | (7.2277)*** | |
| | [3.8576] | | [7.0775]*** | |
| Individual FE | Yes | Yes | Yes | Yes |
| Quarter-year FE | Yes | Yes | Yes | Yes |
| City × Quarter-year FE | No | Yes | No | Yes |
| Observations | 9470 | 9470 | 9470 | 9470 |
| $R^2$ | 0.5997 | 0.6053 | 0.6216 | 0.6270 |

Note: This table reports the value of $\beta_1$ in Eq. (5), linking households' expectation of future housing prices in the HPES with the land auction outcomes. The $\widetilde{\text{Premium}}$ is the centered average land premium within 3-km of individual's home address in each quarter. Land is a dummy indicating whether the individual's home is located within 3-km of any land auction in each quarter. The dependent variable is respondents' expectation of future housing price growth (percent). We control individual and quarter-by-year fixed effects. Standard errors in parentheses are clustered at the household level; standard errors in brackets are heteroskedasticity-robust but not clustered. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

**Table 4**
Effect of developers' nearby land holdings on land premium.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Has nearby land stock | 0.0389 | 0.0475 | 0.0389 | 0.0392 | 0.0507 | 0.0572 | 0.0447 | 0.0339 |
| | (0.0134)*** | (0.0136)*** | (0.0138)*** | (0.0145)** | (0.0140)*** | (0.0164)*** | (0.0157)** | (0.0228) |
| | [0.0039]*** | [0.0038]*** | [0.0037]*** | [0.0037]*** | [0.0048]*** | [0.0049]*** | [0.0049]*** | [0.0056]*** |
| Has nearby land stock | | | | 0.0061 | | 0.0093 | | |
| × log(Area of nearby unsold housing stocks) | | | | | | | | |
| | | | | (0.0029)** | | (0.0035)** | | |
| | | | | [0.0014]*** | | [0.0014]*** | | |
| Has nearby land stock | | | | | 0.0917 | 0.1285 | | |
| × Δ Market share | | | | | | | | |
| | | | | | (0.0270)*** | (0.0295)*** | | |
| | | | | | [0.0168]*** | [0.0176]*** | | |
| Δ Market share | | | | | −0.0218 | −0.0216 | | |
| | | | | | (0.0065)*** | (0.0064)*** | | |
| | | | | | [0.0022]*** | [0.0021]*** | | |
| log(Land area) | −0.0011 | 0.0004 | −0.0020 | −0.0022 | −0.0015 | −0.0019 | −0.0029 | −0.0015 |
| | (0.0011) | (0.0009) | (0.0011)* | (0.0011)* | (0.0012) | (0.0012) | (0.0022) | (0.0011) |
| | [0.0007] | [0.0007] | [0.0007]*** | [0.0007]*** | [0.0007]** | [0.0007]*** | [0.0010]*** | [0.0009] |
| City × Year FE | | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Land type | | | Yes | Yes | Yes | Yes | Yes | Yes |
| Land transfer type | | | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 15,648 | 15,648 | 15,648 | 15,648 | 15,648 | 15,648 | 6703 | 8945 |
| $R^2$ | 0.0096 | 0.0842 | 0.1715 | 0.1731 | 0.1765 | 0.1799 | 0.1720 | 0.1677 |

*Note*: This table reports developers' adjustment in their strategy in land auctions when they have other housing stock near the auctioned land parcel. The dependent variable is defined as the growth rate of the price of the auctioned land relative to the historical price of other lands within a 3-km radius of the auctioned land, subtracting the growth rate of the housing price located within a 3-km radius of the land. The first row reports the impact of whether the developer $d$ has other housing projects near the land on the auction premium. log(Area of nearby unsold housing stocks) measures the total area of unsold nearby housing already under the developer's control. log(Land area) measures the total area supplied by the auctioned land. Δ Market share is calculated as the change in a developer's market share after winning the auction. The market share of the developer after the land auction is calculated as the ratio of (developer's saleable area + newly auctioned area) divided by (total saleable area + newly auctioned area). Land type includes dummy variables representing the designated use of the land. Land transfer type controls the type of auction used to transfer the land parcel. Columns 1 to 3 differ in the type of fixed effects controlled. Columns 4 to 6 differ in the type of interactions included. Columns 7 and 8 report estimation results restricting samples to cities that are in a real-estate market upturn and downturn, respectively. Standard errors in parentheses are clustered at both the developer and city levels, and standard errors in brackets are heteroskedasticity-robust but not clustered. *, **, and *** stand for significance at the 10%, 5%, and 1% levels, respectively.

To unpack why these developers bid more aggressively, columns (4) and (5) introduce two interaction terms.[17] First, we interact the nearby-land indicator with the log total area of unsold nearby housing already under the developer's control. A larger unsold inventory amplifies the auction's impact. We also interact the nearby-land indicator with the change in local market share conferred by the new parcel; the positive coefficient confirms that greater post-auction dominance in local housing market further elevates land prices.[18] Column (6) includes both interactions jointly. Evaluated at sample means, developers with

[17] Both interaction terms are centered, so the linear term reflects the average effect evaluated at the sample means.

[18] The market share of the developer after the land auction is calculated as the ratio of (developer's saleable area + newly-auctioned area) divided by (total saleable area + newly-auctioned area).

**Table 5**
Falsification test for the effect of nearby land holdings on land premium.

| | (1) | (2) |
|---|---|---|
| Has nearby land stock | 0.0392 | 0.0572 |
| | (0.0145)** | (0.0164)*** |
| | [0.0037]*** | [0.0049]*** |
| Has nearby land stock | 0.0004 | 0.0004 |
| × log(Area of sold nearby housing stocks) | | |
| | (0.0008) | (0.0008) |
| | [0.0007] | [0.0007] |
| Has nearby land stock | 0.0061 | 0.0093 |
| × log(Area of nearby housing stocks) | | |
| | (0.0029)** | (0.0035)** |
| | [0.0014]*** | [0.0014]*** |
| Has nearby land stock | | 0.1284 |
| × Δ Market share | | |
| | | (0.0295)*** |
| | | [0.0176]*** |
| Δ Market share | | −0.0216 |
| | | (0.0065)*** |
| | | [0.0021]*** |
| log(Land area) | −0.0022 | −0.0019 |
| | (0.0011)* | (0.0012) |
| | [0.0007]*** | [0.0007]*** |
| City × Year FE | Yes | Yes |
| Land type | Yes | Yes |
| Land transfer type | Yes | Yes |
| Observations | 15,648 | 15,648 |
| $R^2$ | 0.1731 | 0.1799 |

*Note*: This table implements a falsification test adding the interaction term between the log total area of nearby housing stock that has been sold and whether the developer holds land stock in the neighborhood to the specification of columns 4 and 5 in Table 4. All other variables are identical to those defined in Table 4. Standard errors in parentheses are clustered at both the developer and city levels, and standard errors in brackets are heteroskedasticity-robust but not clustered. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

nearby holdings pay an average premium of 5.72 percent. Crucially, a significant share of this effect is attributable to the unsold-stock interaction, highlighting the cross-subsidy channel as the primary driver: developers strategically overbid because a higher auction price bolsters demand—and thus sale prices—for the units they already have on hand in the same neighborhood.

Columns (7) and (8) in Table 4 report estimation results restricting samples to cities that are in a real estate market upturn and downturn, respectively. We find that developers with nearby land stock pay higher prices in a boom market, while the relationship is not statistically significant during market downturns. The evidence is consistent with the observation in the previous section that higher land premiums increase local housing prices during market expansions.

### 5.2. Alternative explanations

Several confounding stories could, in principle, explain why developers with nearby projects bid more aggressively. This subsection shows that none of these rival mechanisms survive a battery of robustness and falsification checks, whereas the patterns are fully consistent with the cross-subsidy channel highlighted in Section 3.

***Falsification test.*** An alternative interpretation of the results is that developers with existing projects in an area, rather than using land auction bids to influence homebuyers' expectations, may hold more optimistic beliefs or benefit from economies of scale, and because of this, they bid more aggressively in subsequent land auctions within the same area. To rule this out, note that our conceptual framework predicts that only *unsold* nearby inventory can motivate a developer to increase the land-auction price: by pushing up the observable price benchmark, the developer simultaneously lifts demand—and thus sale prices—for the units it is still selling. Building on this prediction, we design a falsification test to examine whether developers still pay more for land with the amount of *sold*, rather than unsold, nearby inventory. Because the cross-subsidy channel should no longer operate when nearby properties have been sold, we expect the amount of sold properties to have no effect on the developer's land auction bids.

We interact the nearby-land indicator with the total area of nearby *sold* units and add this to the model in Table 4.[19] The estimates are reported in Table 5. The interaction coefficient is economically and statistically negligible. Once the local pipeline is empty, the incentive to pay more in land auctions disappears. By contrast, the finding cannot be explained by a scale-economy or optimistic-belief interpretation which would predict a premium regardless of a developer's inventory status.

***Financial-capacity mechanism.*** First, it is possible that the observed patterns may simply reflect larger firms, which typically have more housing projects, possessing greater financial resources. Since larger firms usually have better financial resources, they are capable of using higher bidding prices to drive smaller firms out of the market. Therefore, the coefficient $\lambda_1$ would capture the premium that larger firms are willing to pay arising from their financial advantages. To ensure that $\lambda_1$ captures the developers' responses to the spillover effect from the housing market, we implement a robustness check excluding large developers. We define large developers as either listed on China's stock market or listed as the top 100 developers by independent real estate consulting firms. In Column (1) in Table 6, we re-estimate Eq. (6) using a sample excluding large developers defined above. We find that, in this sample of developers with more homogeneous financing abilities, developers with other land stock near the auctioned parcel paid about a 4 percent higher premium. A pure "deep pockets" explanation therefore falls short: the effect of owning nearby housing projects on land premium is present even among mid-sized developers with similar financial capacity.

***Private information.*** Another possibility is that incumbents learn in advance about forthcoming infrastructure developments and bid higher in anticipation of genuine price appreciation. To probe this

[19] The interaction variable has been centered.

**Table 6**
Robustness checks for the effect of nearby land holdings on land premium.

| | Excluding listed and top 100 developers | New schools two years after auction | | New metros two years after auction | |
|---|---|---|---|---|---|
| | | No | Yes | No | Yes |
| | (1) | (2) | (3) | (4) | (5) |
| Has nearby land stock | 0.0362 | 0.0352 | 0.0409 | 0.0301 | 0.0424 |
| | (0.0143)** | (0.0101)*** | (0.0242) | (0.0131)** | (0.0135)*** |
| | [0.0038]*** | [0.0044]*** | [0.0065]*** | [0.0042]*** | [0.0077]*** |
| log(Land area) | −0.0019 | −0.0026 | −0.0013 | −0.0014 | −0.0059 |
| | (0.0012) | (0.0011)** | (0.0025) | (0.0009) | (0.0033)* |
| | [0.0007]*** | [0.0008]*** | [0.0017] | [0.0007]* | [0.0018]*** |
| City × Year FE | Yes | Yes | Yes | Yes | Yes |
| Land type | Yes | Yes | Yes | Yes | Yes |
| Land transfer type | Yes | Yes | Yes | Yes | Yes |
| Observations | 14,553 | 12,503 | 3130 | 13,060 | 2578 |
| $R^2$ | 0.1728 | 0.1532 | 0.3169 | 0.1589 | 0.3240 |

Note: This table reports robustness checks of developers' auction strategies. The first column excludes developers listed in China's stock market or listed as the top 100 developers by independent real-estate consulting firms. Column 2 limits the sample to lands where no new school is built within 3 km of the auctioned land in the two years after the auction. Column 3 limits the sample to lands where a new school is built within 3 km of the auctioned land in the two years after the auction. Column 4 limits the sample to lands where no new metro line is built within 3 km of the auctioned land in the two years after the auction. Column 5 restricts the sample to lands where a new metro line is built within 3 km of the auctioned land in the two years after the auction. We control city-by-year fixed effects, land-use dummies, land-transfer-type dummies, and log(Land area), which measures the total area supplied by the auctioned land. Standard errors in parentheses are clustered at both the developer and city levels, and standard errors in brackets are heteroskedasticity-robust but not clustered. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

idea, we partition the data by whether a new school or metro line opens within 3 km in the two years following the auction and verify whether developers' strategies are similar in both samples. If we find similar responses in both samples, we can rule out the hypothesis that incumbents are more optimistic of future housing prices due to their better access to local infrastructure plan. Columns (2) to (5) in Table 6 report results from these exercises. We find that developers with nearby land stock pay premiums with similar magnitudes, regardless of whether new schools or metro lines are built two years after the auction.

***Mortgage appraisal channel.*** An alternative mechanism is that developers raise appraised values of their existing housing stock to relax buyers' mortgage constraints. Under this "appraisal inflation" channel, high transaction prices on newly auctioned land or nearby properties serve primarily to increase appraisal valuations. This, in turn, enables potential buyers to secure larger mortgages, easing their financing constraints and helping developers sell inventory.

Ideally, we would test this channel using detailed mortgage microdata—for example, by comparing the loan-to-value ratios of buyers from the winning developer's nearby projects after a high-premium land auction with those of other buyers. However, such loan-level data are unavailable in our setting, preventing direct implementation of this test.

Given this data limitation, we evaluate the mechanism's plausibility in light of institutional realities. Mortgage approval depends not only on appraised value but also strictly on borrower income. A buyer with insufficient income cannot obtain a significantly larger loan simply because the appraisal rises. Thus, the appraisal channel can only meaningfully assist a specific subset of buyers: those marginally constrained by the down payment but with sufficiently high income. If high auction prices mainly helped this narrow, cash-constrained group, it would be difficult for the mechanism alone to generate the broad and persistent local price increases we document.

In contrast, the household-beliefs channel we emphasize can operate across a much wider range of buyers. By shifting expectations of future housing prices, high land premiums make current transactions more attractive to both financially constrained and unconstrained households, thereby amplifying the effect of salient public signals.

Furthermore, bidding up land prices is costly for developers. As quantified in Appendix B, paying high land premiums requires substantial upfront investment. If the sole benefit were a temporary easing of financing constraints to accelerate inventory sales, the implied return would appear limited. The magnitude and persistence of the effects we observe are more consistent with a broader shift in beliefs about future prices, which justifies aggressive land bidding than with a reliance on the appraisal-based mortgage channel.

Taken together, these considerations suggest that while appraisal-driven credit easing may play a complementary role, it is unlikely to be the dominant driver of our results. We therefore interpret our findings primarily through the lens of extrapolative household beliefs, while acknowledging that strategic interactions with the mortgage appraisal process may also play a secondary role.

### 5.3. A policy-based test of strategic bidding: Housing price caps

#### 5.3.1. Policy design and implications

Beginning in 2011, several Chinese cities introduced housing sale price caps tied to land auctions. Under this regime, developers could bid freely for land-use rights, but they were required to commit ex ante to a maximum sale price for the housing units built on that parcel. The policy was motivated by local governments' efforts to cool overheated housing markets and curb speculative price growth.

In practice, developers submitted a housing-price commitment letter as part of the auction process. The cap (often referred to as a "guided price") was set by a joint committee involving the housing and urban–rural development, planning, and natural resources departments. Guided prices were determined using the benchmark land price for the parcel's area, the average transaction price of newly built housing in the surrounding market over the preceding three years, and parcel-specific features such as the Floor Area Ratio (FAR) and local infrastructure conditions. The resulting guided price was publicly announced during the notice period, making the price ceiling salient and commonly known to both developers and potential homebuyers. Although formal documents did not always state that the cap must lie below market-clearing expectations, local officials widely understood the policy's intent as limiting further price run-ups and reducing speculative demand.

A telling example comes from Hangzhou's first auction after the price cap policy implemented in 2019. For a parcel in the city with a burgeoning tech sector, the regulated new home price was fixed at 34,600 RMB per square meter. This stood in stark contrast to the surrounding secondary market, where similar apartments routinely traded above 40,000 RMB per square meter. Crucially, this was not an

**Table 7**
Impact of nearby land holdings on land premium, by city's housing price cap policy.

| | No price cap | Price cap | Two years after price cap |
|---|---|---|---|
| | (1) | (2) | (3) |
| Has nearby land stock | 0.0660 | −0.0024 | 0.0007 |
| | (0.0178)*** | (0.0063) | (0.0080) |
| | [0.0047]*** | [0.0054] | [0.0058] |
| log(Land area) | −0.0030 | −0.0016 | −0.0018 |
| | (0.0020) | (0.0011) | (0.0010)* |
| | [0.0009]*** | [0.0010]* | [0.0011] |
| City × Year FE | Yes | Yes | Yes |
| Land type | Yes | Yes | Yes |
| Land transfer type | Yes | Yes | Yes |
| Observations | 7000 | 8648 | 7384 |
| $R^2$ | 0.2191 | 0.1598 | 0.1598 |

Note: This table reports developers' adjustment in their strategy in land auctions in response to the spillover effects separately for cities with and without the housing price limit policy. The dependent variable is defined as the growth rate of the price of auctioned land relative to the historical price of other lands within a 3-km radius of the auctioned land, subtracting the growth rate of the housing price located within a 3-km radius of the land. The first row reports the impact of whether the developer has other housing projects or land stock near the land on the auction premium. The first column reports the estimates based on the sample of cities that did not implement the housing price limit policy. The second column reports the estimates based on the sample of cities that implemented the housing price limit policy. The third column excludes observations within two years after the policy implementation to avoid contamination from projects that were already on sale before the price cap. We control city-by-year fixed effects, land-use dummies, land-transfer-type dummies, and log(Land area), which measures the total area supplied by the auctioned land. Standard errors in parentheses are clustered at both the developer and city levels, and standard errors in brackets are heteroskedasticity-robust but not clustered. *, **, and *** stand for significance at the 10%, 5%, and 1% levels, respectively.

isolated case but rather represented the typical outcome across virtually all cities that adopted the policy.[20]

For our purposes, the key implication is that the policy provides an externally imposed and publicly observed ceiling on near-term selling prices for units built on newly acquired land. This feature helps us isolate the strategic channel emphasized in our mechanism. Our empirical design focuses on a within-market comparison: we contrast bids by developers who hold nearby land/housing stocks with bids by developers who do not. To the extent that the policy affects the entire local market (e.g., by changing aggregate beliefs), such common effects are differenced out in this comparison. The remaining differential response identifies how price caps attenuate the cross-subsidy motive: when sale prices on new projects are capped and publicly fixed at the time of auction, incumbents have less scope to rationalize aggressive bidding for additional land using higher expected selling prices from their existing nearby projects.

Consistent with this prediction, we estimate Eq. (6) separately for cities that adopted price caps and for those that did not. Developers with nearby land pay a 6.6% premium in unconstrained cities, whereas the effect becomes close to zero and statistically insignificant in cities where caps are in force (Table 7, columns (1) and (2)).

Second, for a land parcel at auction, its nearby homes may also be subject to price caps if they were constructed on land acquired after the implementation of the house price cap policy. This may further reduce incumbent developers' incentive to use land auction bids to influence demand for their nearby properties. Furthermore, land and housing transactions involve complex decisions, and the policy's effects may be time-varying as market participants process the information. Considering this mechanism, we exclude land auctions within two years following the policy's adoption. The results confirm that incumbent developers no longer systematically bid higher in this scenario (Table 7 Column (3)).

### 5.3.2. Addressing endogeneity in policy timing

We now turn to the potential endogeneity of policy adoption. Cities that implemented price caps were typically those with rapidly rising housing prices and optimistic developer sentiment. If these underlying trends also affect developers' bidding behavior, our estimates could confound the causal impact of the policy with pre-existing differences in local fundamentals.

We address these identification challenges through several features of our research design. First, our empirical strategy incorporates city by year fixed effects, which absorb time-varying city-level unobservables such as local economic conditions, housing demand shocks, and administrative preferences that could influence both policy adoption and auction outcomes. This mitigates concerns about bias from time-varying city-level endogeneity.

Second, we provide complementary evidence using within-city variation around the timing of adoption. Specifically, we compare parcels won by developers that already hold nearby land with parcels won by developers without such holdings, and examine how this gap evolves around the adoption of the price-cap policy. The identifying assumption is that, absent the policy, the difference in bidding premiums between these two groups would have followed parallel trends within the same city. This design nets out time-varying city-level factors, such as housing price dynamics and concurrent local policies, that affect all developers operating in the city. The estimating equation is:

$$y_{ljt} = \sum_{n \neq -1} \beta_1^n * \text{PriceCap}_{jt}^n * Z_l + \beta_2 Z_l + \beta_3 X_l + w_{jt} + \epsilon_{ljt}, \tag{7}$$

where the dependent variable $y_{ljt}$ is the land premium of parcel $l$ in city $j$ and year $t$; $\text{PriceCap}_{jt}^n$ is an event-time indicator equal to one if $t$ is $n$ years relative to the year in which city $j$ first adopted the price-cap policy (with $n = -1$ omitted as the reference year); and $Z_l$ is an indicator for whether the winning developer of parcel $l$ holds other land within the specified nearby radius. $X_l$ includes parcel-level controls, and $w_{jt}$ denotes city-by-year fixed effects; the remaining variables follow the definitions in Eq. (6).[21]

The coefficients $\beta_{1\,n\neq-1}^n$ trace out a dynamic profile of the policy's impact on the premium gap between developers with and without nearby land holdings. Concretely, each $\beta_1^n$ measures how the difference in land premiums between the two groups in event year $n$ changes relative to that same difference in the reference year $n = -1$. Thus, the post-policy coefficients ($n \geq 0$) capture how the price-cap policy differentially alters the bidding behavior of developers with nearby land.

The estimated $\beta_1^n$ coefficients are displayed in Fig. 11. We find a pronounced decline thereafter the policy: developers with nearby land holdings bid significantly lower premiums after the price-cap policy takes effect, relative to developers without nearby land. This within-city event-study evidence strengthens the interpretation that the policy disrupts the strategic cross-subsidy motive.

Finally, if endogenous policy adoption were a major confounder, one would expect the policy to have little measurable effect on premiums, as it would be introduced precisely where high premiums reflected strong fundamentals. Yet our results show a clear reduction in premiums for developers with nearby landholdings, a pattern inconsistent with pure selection and supportive of a causal policy effect.

In sum, while policy adoption may not be random, our research design and empirical results provide robust evidence that the price-cap policy directly alters developers' bidding behavior, consistent with our theoretical mechanism.

[20] Newspaper coverage reported widespread "first vs second-hand price inversion" under price caps in multiple cities. See https://paper.people.com.cn/zgcsbwap/html/2021-01/04/content_2027172.htm.

[21] The linear term $\text{PriceCap}_{jt}^n$ is absorbed by the city-by-year fixed effects.

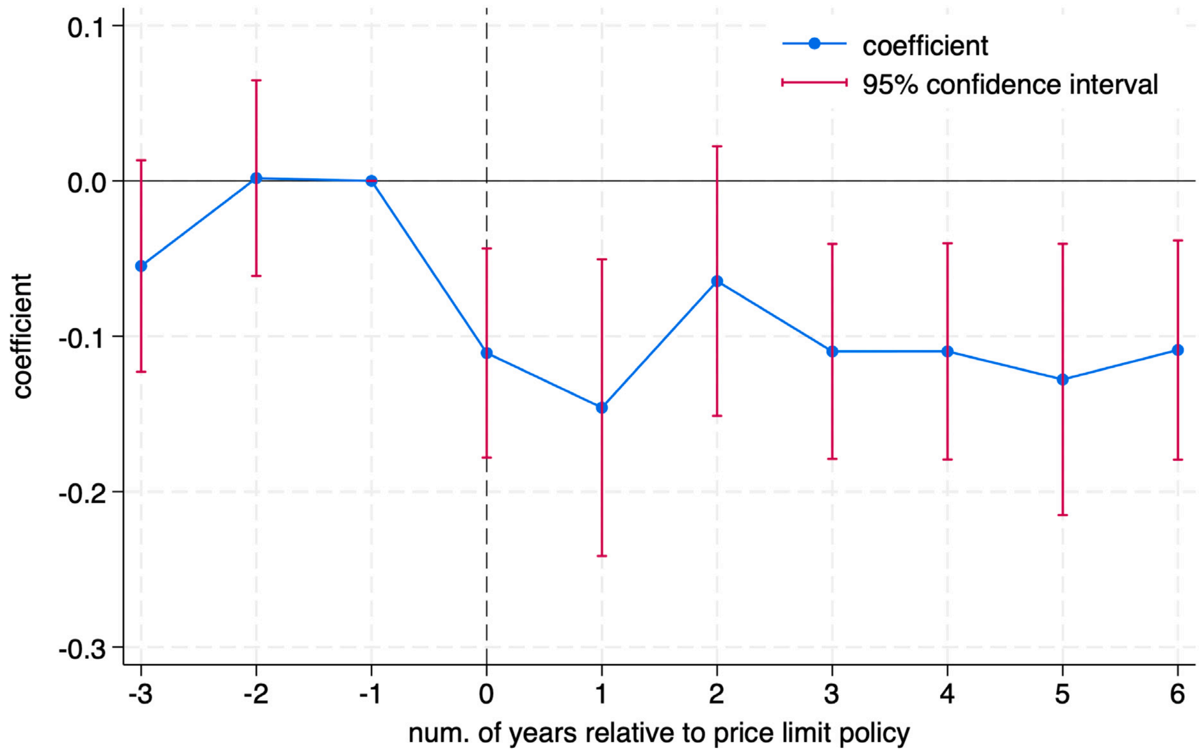



**Fig. 11.** Dynamic effect of the housing price limit policy on land premiums: difference between developers with and without nearby land holdings.
*Note*: This figure plots the coefficients $\hat{\beta}_1^n$ from estimating Eq. (7). The dependent variable $y_{ljt}$ is the land premium, defined as land price growth around the auctioned parcel net of nearby housing price growth (both measured within a 3-km radius. The vertical dashed line marks policy adoption ($n = 0$). Points are coefficient estimates and bars are 95% confidence intervals.

### 5.4. Discussions

Synthesizing the direct evidence with exhaustive robustness checks, we demonstrate that developers strategically inflate land premiums to capitalize on homebuyers' extrapolative expectations. Developers with existing land holdings near an auctioned parcel pay higher premiums than other bidders. This strategic bidding is primarily driven by the scale of developers' unsold local inventory, consistent with a cross-subsidization motive whereby developers recoup land acquisition costs through subsequent price appreciation of their existing unsold holdings. The premium persists among mid-sized developers, is robust to the future construction of local infrastructure, and vanishes when interacting with sold (rather than unsold) inventory. This pattern is inconsistent with mechanisms based on "deep pockets", information advantages, or economies of scale. A quasi-experimental policy test provides decisive evidence: in cities that cap future housing prices, severing the link between land prices and household expectations, the premium decreases substantially. The evidence collectively indicates that developers engage in strategic bidding, inflating land prices to signal future housing appreciation and thereby boost the value of their unsold nearby stock.

To illustrate the profitability of this strategy, we perform a back-of-the-envelope calculation based on the land auctions presented in Fig. 3. As discussed in Section 2.2, Fig. 3 highlights the spatial relationship between two land parcels in Guangzhou, acquired in 2014 and 2016 by the same developer. The developer paid 3.90 billion yuan for the second land parcel, a price significantly above its market value. Our calculation indicates that the substantial premium paid for the second parcel yielded a net gain of 0.26 billion yuan, approximately 6.7% of the overall investment cost.[22] This case study provides a concrete example of the interaction between strategic land auction dynamics, resultant housing market price movements, and developer profitability.

The documented strategic behavior has not gone unnoticed by regulators. In response, local governments in China have experimented with various policy measures. A prominent example, rolled out from 2016 onward, was the imposition of price ceilings on land auctions, often combined with lotteries to allocate parcels once the ceiling was reached. These interventions were explicitly designed to prevent bidding wars and the artificial inflation of land values. However, the broader market context has shifted dramatically. Beginning in 2021, the housing market entered a pronounced downturn, which sharply reduced local governments' revenues from land sales. By 2023, total land sale revenue had declined to 4203 billion yuan—a 45% drop compared to the 2019 peak. Faced with severe fiscal strain due to their heavy reliance on land-based revenues, many local governments have been forced to reconsider these market-cooling interventions. Consequently, a growing number of cities have removed land price ceilings altogether in an attempt to reinvigorate developer interest and stabilize revenues. This policy reversal, while fiscally motivated, risks recreating the very conditions that enable the strategic impact we identify, presenting a trade-off between short-term fiscal stability and long-term market efficiency and equity.

Overall, these shifting policy landscapes and the strategic behaviors of developers highlight the intricate interplay between government

[22] Appendix C provides the detailed calculations supporting the estimated net gain from the high land premium.

intervention, firm strategy, and household belief formation in the property market. Understanding these dynamics is crucial for designing policies that not only stabilize prices and secure public revenues but also promote fairness, efficiency, and long-term welfare. One immediate implication is that revenue-maximizing local governments may find it advantageous to sell land to developers who own nearby parcels, as this can trigger higher bids. However, from a broader social welfare perspective, this approach risks fueling price bubbles through household extrapolation, undermining housing affordability, and exacerbating price volatility. Therefore, policymakers must carefully weigh the trade-offs between fiscal incentives and the potential for strategic developer behavior when designing land auction mechanisms.

## 6. Concluding remarks

This study has provided a comprehensive analysis of the impact of land auctions on housing market dynamics in China. Our findings reveal significant spillover effects from land auctions to the housing market, particularly in areas proximate to auctioned land parcels. Developers' strategies are evidently influenced by their existing housing stock near auctioned land parcels, which affects the bidding behavior and the resulting auction premiums. We observed that in cities with a more pronounced housing market upturn, the effect of land auctions on housing prices and demand was more substantial. Conversely, during downturns, the influence of land auctions appeared to be mitigated.

The research also highlights the role of local government policies in shaping market dynamics. In cities with fixed housing price policies, developers' strategies differ notably from those in cities without such policies. This finding indicates the significant impact of regulatory environments on market behavior and outcomes.

Overall, this study contributes to the understanding of the interplay between land auctions and housing markets, offering insights into developer strategies and the effects of policy interventions. It underscores the importance of considering the localized nature of real estate markets and the varying impact of external factors across different economic cycles. The implications of this study are valuable for policymakers, developers, and investors in formulating strategies and policies that align with market dynamics and trends.

## CRediT authorship contribution statement

**Zhe Liu:** Writing – original draft, Formal analysis, Data curation. **Wei Shi:** Writing – review & editing, Writing – original draft, Conceptualization. **Yan Song:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Weizeng Sun:** Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jue.2026.103864.

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